Atlas Educational Library Content Requirements Map: A Physics-First System Communication Framework

Section 1: Philosophical and Physics-First Framework

Transitioning the domestic heating sector from high-carbon fossil fuels to low-carbon, low-temperature systems requires a profound shift in communication strategy. Historically, domestic space heating in the United Kingdom relied on high-temperature gas or oil boilers operating with flow and return temperatures of 70°C to 82°C. These systems delivered brief, high-intensity thermal bursts to relatively small radiators. This allowed installers to overlook precise hydraulic design, room-by-room heat loss calculations, and water quality standards.

In contrast, modern low-temperature systems—such as air-to-water heat pumps and high-efficiency condensing gas boilers—rely on continuous, low-intensity heat transfer. To maintain thermodynamic efficiency, these systems operate at lower flow temperatures, typically between 35°C and 55°C.

The physical principles of low-temperature heat distribution are governed by the fundamental heat transport equation:

where Q is the thermal output (W), $\dot{m}$ is the fluid mass flow rate (kg/s), c_p is the specific heat capacity of the heat transfer fluid ($\text{J/kg}\cdot\text{K}$), and ΔT is the temperature differential across the emitter (K).

When the design flow temperature is reduced to optimize the Seasonal Coefficient of Performance (SCOP) of a heat pump :

the mean temperature difference between the radiator surface and the room air decreases. To maintain identical thermal comfort (Q) without increasing the emitter surface area, the mass flow rate ($\dot{m}$) must be significantly increased, or the design temperature differential (ΔT) must be narrowed.

Increasing the mass flow rate through legacy hydraulic networks raises the fluid velocity (v). Under the Darcy-Weisbach equation:

where ΔP is the frictional pressure drop (Pa), f is the friction factor, L is the pipe length (m), D is the internal diameter (m), and ρ is the fluid density (kg/m^3), the hydraulic resistance scales quadratically with velocity.

If water velocity exceeds the critical thresholds of $1.0\text{ to }1.5\text{ m/s}$, the system is prone to erosive turbulence, cavitation, and significant pipework noise.

The Atlas Educational Library serves as a critical translation layer. It bridges the gap between thermodynamic realities and the historical assumptions of homeowners, surveyors, and engineers. Historically, customers evaluated a system’s performance by touching a "scalding hot" radiator.

The customer communication library must shift this expectation toward continuous, low-temperature thermal comfort.

Furthermore, from December 2024, the Microgeneration Certification Scheme (MCS) references BS EN 12831-1:2017 under the MIS 3005-D standard for room-by-room heat loss calculations.

This replaces the outdated BS EN 12831:2003 standard, mandating precise fabric and ventilation calculations.
Similarly, British Standard BS 7593:2019+A1:2024 mandates inline magnetic system filters, regular water quality testing, and chemical water treatment.
The Atlas system must explain these regulatory requirements not merely as compliance burdens, but as essential steps to protect warranties and optimize efficiency.

Section 2: Categorized Map of the Atlas Educational Library

The Atlas Educational Library is organized into ten distinct research domains. These cover every physical, hydraulic, and behavioral aspect of modern domestic thermal networks.

Domain A: System Identity and Topology

Explains the layout and components of the home's heating and hot water network. This domain prepares the user for hydraulic conversions recommended by the Atlas engine.

```
[CON_A01] Open-Vented to Sealed System Conversion ----- Priority: MVP
[CON_A02] Low-Loss Headers and Buffer Vessels --------- Priority: MVP
[CON_A03] Combi Boilers vs. System Boilers ------------ Priority: MVP
[CON_A04] Heat-Only (Regular) Boilers ----------------- Priority:
Phase 2
[CON_A05] Unvented Hot Water Cylinders ---------------- Priority: MVP
[CON_A06] Vented Cylinders and Header Tanks ----------- Priority:
Phase 2
[CON_A07] Thermal Stores ------------------------------ Priority:
Later
[CON_A08] Smart Hot Water Cylinders ------------------- Priority:
Phase 2
[CON_A09] Heat Pumps (ASHP & GSHP) -------------------- Priority: MVP
[CON_A10] Hybrid Systems (Gas-Heat Pump) -------------- Priority:
Phase 2
[CON_A11] Primary vs. Secondary Circuits -------------- Priority: MVP
[CON_A12] Flow and Return Temperatures ---------------- Priority: MVP
[CON_A13] Circulator Pumps and Zone Valves ------------ Priority:
Phase 2
[CON_A14] Expansion Vessels --------------------------- Priority: MVP
[CON_A15] Feed and Expansion (F&E) Tanks -------------- Priority:
Phase 2
[CON_A16] Hydraulic Separation Principles ------------- Priority:
Later
```

Domain B: Boiler and Heat-Source Behaviour

Focuses on operational dynamics, modulation, and physical vulnerabilities of fuel-burning appliances. It emphasizes why modern condensing boilers require low return temperatures to

achieve their rated efficiency.

```
Boiler Sizing: Why Bigger Is Not Better ----- Priority: MVP
Gas Pipe Sizing and Pressure Drop ----------- Priority: MVP
System Pressure and the Filling Loop -------- Priority: MVP
Condensing Operation and Return Temps ------- Priority: MVP
Boiler Modulation vs. Short Cycling --------- Priority: Phase 2
Burner Ramp Rates and Anti-Cycle Delays ----- Priority: Later
Hot Water Priority (Diverter Valves) -------- Priority: MVP
Combi Plate Heat Exchanger Scale ------------ Priority: Phase 2
Cold Water Sandwich Effect ------------------ Priority: Phase 2
Flue Siting and Safety Clearance ------------ Priority: MVP
Condensate Pipe Routing and Freezing -------- Priority: MVP
Gas Rate Verification ----------------------- Priority: Phase 2
```

Domain C: Hot Water Systems

Addresses stored hot water generation, distribution hydraulics, and sanitary safety. It balances user expectations of high flow rate with thermodynamic recharge limits.

```
[CON_C01] Unvented Cylinder Safety and G3 Code -------- Priority: MVP
[CON_C02] Hot Water: Pressure vs. Storage Volume ------ Priority: MVP
[CON_C03] Smart Tariff Hot Water Scheduling ----------- Priority: MVP
[CON_C04] Mains-Pressure Hot Water Delivery ----------- Priority: MVP
[CON_C05] Simultaneous Outlet Performance ------------- Priority: MVP
[CON_C06] Cylinder Recovery and Heat Up Times --------- Priority:
Phase 2
[CON_C07] Standing Heat Losses ------------------------ Priority:
Phase 2
[CON_C08] Cylinder Stratification --------------------- Priority:
Later
[CON_C09] Top-Down Smart Cylinder Heating ------------- Priority:
Phase 2
[CON_C10] Immersion Heaters and Emergency Backup ------ Priority: MVP
[CON_C11] Secondary Circulation and Dead Legs --------- Priority:
Phase 2
[CON_C12] Thermostatic Mixing Valves (TMV) ------------ Priority:
Phase 2
[CON_C13] Tundish and Discharge Pipe Routing ---------- Priority: MVP
[CON_C14] Legionella Protection and Pasteurization ---- Priority: MVP
```

Domain D: Water Supply and Hydraulics

Explains the mechanics of fluid flow, pipework restrictions, and pressure management, establishing the thermodynamic reality that flow rate is governed by system friction.

```
Static Water Pressure vs. Dynamic Flow ------ Priority: MVP
Accumulators vs. Booster Pumps -------------- Priority: MVP
Microbore Pipe Hydraulic Limitations -------- Priority: MVP
```

```
Hydraulic Balancing and Lockshields --------- Priority: MVP
Lead / Common Water Main Restrictions ------- Priority: MVP
Full-Bore Flow Testing ---------------------- Priority: MVP
Friction Loss and Pipe Bore Dynamics -------- Priority: Phase 2
Pump Head and System Resistance ------------- Priority: Phase 2
Cavitation and Velocity Noise --------------- Priority: Phase 2
Air Locks and De-aeration ------------------- Priority: MVP
Automatic Bypass Valves (ABVs) -------------- Priority: MVP
Water Regulations and WRAS Approvals -------- Priority: Phase 2
```

Domain E: Radiators, Emitters, and Heat Loss

Explains the thermodynamics of heat emission. It helps customers understand that radiator surface area is the primary lever for reducing flow temperatures and saving energy.

```
[CON_E01] Radiator Surface Area and Output ------------ Priority: MVP
[CON_E02] Warm Radiators in Low-Temp Systems ---------- Priority: MVP
[CON_E03] Room-by-Room Heat Loss Calculations --------- Priority: MVP
[CON_E04] Building Fabric vs. Ventilation Heat Loss --- Priority:
Phase 2
[CON_E05] Radiator Types (Type 11, 21, 22, 33) -------- Priority: MVP
[CON_E06] Convector Fins and Air Convection ----------- Priority:
Phase 2
[CON_E07] Furniture Blocking Radiator Emitters -------- Priority: MVP
[CON_E08] Towel Rails: Purpose and Limitations -------- Priority: MVP
[CON_E09] Underfloor Heating (UFH) Thermodynamics ----- Priority: MVP
[CON_E10] Radiator Upgrade vs. Boiler Upsizing -------- Priority: MVP
```

Domain F: Cleaning, Water Quality, and Protection

Highlights chemical water treatment and physical separation. It explains how system cleanliness prevents premature component failures and maintains long-term efficiency.

```
[CON_F01] Powerflush vs. Chemical Flush --------------- Priority: MVP
[CON_F02] Why Flushing Protects Your New Boiler ------- Priority: MVP
[CON_F03] When Powerflushing Is Risky ----------------- Priority: MVP
[CON_F04] Magnetic Filters and Sludge Protection ------ Priority: MVP
[CON_F05] Chemical Inhibitors and Biocides ------------ Priority:
Phase 2
[CON_F06] Limescale and Water Softeners --------------- Priority:
Phase 2
[CON_F07] Corrosion, Sludge, and Black Magnetite ------ Priority: MVP
[CON_F08] BS 7593 Water Quality Standards ------------- Priority: MVP
[CON_F09] Dirt and Air Separators --------------------- Priority:
Phase 2
[CON_F10] Water Quality Test Certificates ------------- Priority: MVP
```

Domain G: Controls and Customer Behaviour

Focuses on the interfaces and behavioral habits that unlock modern heating system efficiency, shifting users from manual on/off control to automated optimization.

```
[CON_G01] Weather and Load Compensation --------------- Priority: MVP
[CON_G02] Thermostat Basics vs. Dial-Up Fallacies ----- Priority: MVP
[CON_G03] Smart TRVs vs. Whole-House Balancing -------- Priority: MVP
[CON_G04] Setback Temperatures vs. Letting Rooms Cold - Priority: MVP
[CON_G05] Manual Overrides and Holiday Modes ---------- Priority:
Phase 2
[CON_G06] Wi-Fi Failure and App-Free Operation -------- Priority: MVP
[CON_G07] The "Hands Off" Heating Rule ---------------- Priority: MVP
```

Domain H: Heat Pumps

Explains the mechanics, operation, and efficiency profiles of low-temperature refrigeration cycles.

```
[CON_H01] Heat Pump Defrost Cycles and Steam ---------- Priority: MVP
[CON_H02] Refrigeration Cycle and COP ----------------- Priority:
Phase 2
[CON_H03] Seasonal Efficiency (SCOP/SPF) -------------- Priority:
Phase 2
[CON_H04] Continuous Low-Temperature Running ---------- Priority: MVP
[CON_H05] Outdoor Fan Noise and Siting ---------------- Priority: MVP
[CON_H06] Condensate Drainage Requirements ------------ Priority: MVP
[CON_H07] Electricity vs. Gas: Operating Costs -------- Priority: MVP
[CON_H08] Myth: "Heat Pumps Don't Work in Cold" ------- Priority: MVP
[CON_H09] Myth: "You Need Perfect Insulation" --------- Priority: MVP
```

Domain I: Installation and Handover

Covers property preparation, physical disruption, statutory certifications, and onboarding requirements for a successful system transition.

```
[CON_I01] Siting Cylinders and Cupboard Space --------- Priority: MVP
[CON_I02] Loft Tank Decommissioning & Space Reclaim --- Priority: MVP
[CON_I03] Commissioning, Balancing, and Benchmark ----- Priority: MVP
[CON_I04] External Wall Drilling and Making Good ------ Priority:
Phase 2
[CON_I05] Home Isolation (Gas, Water, Power) ---------- Priority: MVP
[CON_I06] Warranty Conditions & Annual Servicing ------ Priority: MVP
 What Customers Should Expect ----- Priority: MVP
```

Domain J: Future-Ready and Home Energy

Links wet heating systems to the broader home energy ecosystem, positioning the thermal network as a dynamic battery.

```
[CON_J01] Heat Pump Readiness: $55^\circ\text{C}$ Retrofit - Priority:
MVP
[CON_J02] Solar PV and Diverters ---------------------- Priority:
Phase 2
[CON_J03] Dynamic Tariffs and Battery Storage --------- Priority:
Phase 2
[CON_J04] Fabric-First Insulation Roadmap ------------- Priority:
Phase 2
```

Section 3: Deep-Dive Concept Blueprints

Focus 1: Powerflush vs. Chemical Flush

- **Concept ID:** CON_F01
- **Plain-English title:** System Deep Clean (Powerflush) vs. Standard Flush (Chemical Wash)
- **Category:** Cleaning, Water Quality, and Protection
- **Customer question it answers:** "Why does the quotation list a Powerflush alongside a Chemical Flush, and what makes the Powerflush more costly?"
- **Why it matters:** Silt and black iron oxide build up in hydronic heating systems, creating cold spots and reducing efficiency. While a standard chemical wash relies on the existing system pump, a Powerflush uses specialized external machinery. Choosing the incorrect cleaning method can leave debris behind, which can clog narrow heat exchangers and void manufacturer warranties.
- **Atlas trigger conditions:** The system detects an existing radiator circuit over 5 years old with visible rust indicators, or the surveyor records high water turbidity on site.
- **Engine facts needed:** Existing system volume (L), pipework configuration, radiator count, and water turbidity measurements.
- **Customer wording:** "A standard chemical flush uses your system's existing pump to circulate a mild cleaner, making it best for newer or well-maintained systems. A Powerflush is a deep clean that connects a high-power external machine. This machine pumps cleaning chemicals through each radiator individually, using high-speed, low-pressure water to strip away stubborn, baked-on sludge."
- **One-line summary:** A standard wash dissolves light debris using your system's existing pump, while a deep Powerflush uses an external machine to physically strip away stubborn sludge, radiator by radiator.
- **What you may notice:** An engineer will bring a large, wheeled pumping machine into the home, connect temporary hoses to the system, and vibrate each radiator with a rubber mallet to loosen hardened rust.
- **What stays familiar:** Radiators and pipework remain in place. The water inside is replaced with fresh, treated water.
- **What not to worry about:** The powerflushing machine uses low pressure (under 2 bar) to protect system joints, relying on flow speed rather than high pressure to clear debris.
- **Common misunderstanding:** "The high pressure of a Powerflush will rupture healthy

pipes."

- **Reality:** Powerflushing relies on high water velocity (flow speed), not high static pressure. The pressure is kept low to safeguard joints while fast-moving water sweeps debris away.
- **Dangerous oversimplification to avoid:** Describing both flushes as "putting chemicals in the pipes" can lead customers to believe they are paying more for the same service.
- **Best analogy families:** Kitchen/cooking.
- **Where analogy works:** A standard chemical wash is like rinsing a dirty lasagna pan with warm soapy water. A Powerflush is like using a high-pressure kitchen scrub brush to clean baked-on grease from each corner.
- **Where analogy breaks:** Pan cleaning is visible, whereas hydronic pipes are concealed under floorboards, making fluid dynamics and velocity hard to see directly.
- **Best graphic:** A comparative split-screen diagram. The left side shows a radiator running a standard chemical wash with sludge remaining in the corners. The right side shows a Powerflush machine connected, with arrows showing high-velocity water lifting sludge from the bottom of the radiator.
- **Best animation:** A 2D animation showing a cross-section of a radiator. Watch black sludge slowly dissolve and sweep away as velocity increases and the outer shell is tapped, leaving clean metal behind.
- **Best print sheet:** A double-sided A5 card with a visual comparison table (Time, Cost, Sludge Removal Rating, and Suitability) on the front, and a checklist of "How to tell which one you need" on the back.
- **Portal treatment:** An interactive diagnostic card. Users click on symptoms (e.g., "Radiator cold at the bottom," "Boiler makes kettling noises") to see which flush matches their system's needs.
- **PDF treatment:** A dedicated technical insert page in the quotation package that outlines the engineering recommendation based on the surveyor's on-site water test.
- **Engineer handover treatment:** A commissioning sheet where the engineer records pre-flush and post-flush water turbidity readings, total flush time, and the volume of water used.
- **Technical appendix:** Detailed fluid dynamic requirements. A minimum flushing velocity of 1.5\text{ m/s} is required in each circuit to mobilize magnetite particles, as specified in BS 7593:2019.
- **Accessibility notes:**
    - *Dyslexia:* Use high-contrast visual status indicators (Red/Green) rather than text-heavy tables.
    - *ADHD:* Break the explanation into bite-sized tabs with interactive elements.
    - *Low literacy:* Avoid technical jargon, using terms like "deep clean" instead.
    - *Technophobia:* Focus on the physical cleaning machinery and clear outcomes.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
    - *Motion sensitivity:* Disable autoplay on the animation, using a play/pause toggle instead.
- **Priority:** MVP
- **Related concepts:** CON_F02, CON_F03, CON_F07
- **Suggested diagram ID:** DIA_F01_PF_VS_CF
- **Suggested animation ID:** ANI_F01_PF_VELOCITY
- **Suggested print card ID:** CRD_F01_FLUSH_COMP

Focus 2: Why Flushing Protects Your New Boiler

- **Concept ID:** CON_F02
- **Plain-English title:** Why Clean Water Protects Your New Boiler and Warranty
- **Category:** Cleaning, Water Quality, and Protection
- **Customer question it answers:** "My old heating system worked fine. Why must I pay to flush the system when installing a new boiler?"
- **Why it matters:** Modern, high-efficiency boilers feature compact heat exchangers with narrow water channels. Leaving old sludge in the system can clog these channels within six days, reducing efficiency and potentially invalidating your warranty.
- **Atlas trigger conditions:** Every boiler replacement recommendation generated by the Atlas sizing engine.
- **Engine facts needed:** New boiler heat exchanger material (aluminum vs. stainless steel), old system pipework material, and estimated age of existing radiators.
- **Customer wording:** "Modern, high-efficiency boilers are designed with ultra-narrow internal channels to heat water as quickly as possible. If you don't flush your system, old sludge and rust from your existing radiators will wash directly into your new boiler, clogging these channels, reducing efficiency, and potentially causing a breakdown."
- **One-line summary:** Flushing protects your new boiler by preventing old, circulating radiator rust from clogging its narrow, high-efficiency heating channels.
- **What you may notice:** The system runs quieter, radiators heat up more evenly, and your energy bills are lower because the boiler can transfer heat easily.
- **What stays familiar:** Your pipework and radiator layouts do not change, but the water inside them is clean and clear.
- **What not to worry about:** The cleaning process is completely self-contained. The engineer safely drains and disposes of all sludge away from your living areas.
- **Common misunderstanding:** "Flushing is just an optional, extra service plumbers use to increase the bill."
- **Reality:** Flushing is a mandatory requirement under UK Building Regulations (Part L) and British Standards (BS 7593). Skipping it can invalidate your manufacturer’s warranty on day one.
- **Dangerous oversimplification to avoid:** Describing flushing as "just cleaning the radiators" ignores its primary goal: protecting the new, expensive heat exchanger inside the boiler.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing your old pipework to a dusty road and your new boiler to a high-performance sports car. If you drive the car on a dusty road without cleaning it first, the air and oil filters will clog immediately, destroying the engine.
- **Where analogy breaks:** Cars use air and oil filters that are easily swapped, whereas a clogged primary boiler heat exchanger often requires replacing the entire unit.
- **Best graphic:** A cross-section diagram comparing an old, wide cast-iron heat exchanger (unaffected by minor sludge) with a modern, narrow stainless-steel coil heat exchanger clogged with black magnetite.
- **Best animation:** An animation showing clear water flowing through a clean boiler coil, followed by a surge of black sludge from an old radiator that wedges inside the coil, causing it to overheat and error out.
- **Best print sheet:** An A4 "Warranty Protection Certificate" template. This explains why

water quality is a prerequisite for manufacturer coverage, using official guidelines from major brands.
- **Portal treatment:** A visual "Sludge Risk Meter" that calculates the customer's risk profile based on system age, helping them understand why the system wash was recommended.
- **PDF treatment:** A highlighted sidebar in the quote booklet: "Manufacturer Warranty Guarantee: Why Water Treatment is Mandatory."
- **Engineer handover treatment:** Integration with the Benchmark commission log, requiring chemical inhibitor level checks and a signed water quality declaration.
- **Technical appendix:** Detailed analysis of heat transfer degradation. A 1\text{ mm} buildup of magnetite on heat exchanger surfaces can reduce heat transfer efficiency by up to 15\%, leading to elevated flue gas temperatures and structural thermal stress.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of pipe diameters to help users grasp the concept quickly.
    - *ADHD:* Use bright, distinct callout boxes to highlight warranty risks.
    - *Low literacy:* Focus on the term "boiler health protection" instead of "heat exchanger fouling."
    - *Technophobia:* Reassure the user that the automated system handles these protections.
    - *Visual impairment:* Ensure all image-heavy pages are readable by screen-reading software.
    - *Motion sensitivity:* Use static diagrams with clear annotations.
- **Priority:** MVP
- **Related concepts:** CON_F01, CON_F04, CON_F08.
- **Suggested diagram ID:** DIA_F02_HX_CLOGGING
- **Suggested animation ID:** ANI_F02_SLUDGE_TRAVEL
- **Suggested print card ID:** CRD_F02_WARRANTY_PROT

Focus 3: When Powerflushing is Risky

- **Concept ID:** CON_F03
- **Plain-English title:** Balancing Powerflushing Power with Old System Care
- **Category:** Cleaning, Water Quality, and Protection
- **Customer question it answers:** "I've been told that a Powerflush might cause leaks in my older heating system. Is this true, and what are the risks?"
- **Why it matters:** In systems with thin, corroded radiators or degraded joints, the physical velocity of a Powerflush can dislodge the very rust plugs that are currently holding a leak at bay, resulting in water damage.
- **Atlas trigger conditions:** Property features steel panel radiators that are more than 20 years old, visible external corrosion on joints, or legacy iron pipework.
- **Engine facts needed:** Radiator age, pipe material, history of previous leaks, and presence of microbore pipework.
- **Customer wording:** "Over decades, rust can eat away at steel radiators from the inside out, creating tiny pinholes that are temporarily plugged by sludge. A high-speed Powerflush clears this sludge away, which can occasionally reveal these pre-existing pinholes and cause a small leak."
- **One-line summary:** Powerflushing does not damage healthy pipes, but clearing out old

sludge can occasionally reveal pre-existing rust holes, leading to small leaks in fragile systems.

- **What you may notice:** The engineer will pressure-test your system before starting and inspect older radiators for damp spots or weeping joints.
- **What stays familiar:** The system layout remains intact, but the engineer may recommend replacing particularly fragile radiators instead of flushing them.
- **What not to worry about:** If a leak does occur, engineers carry leak sealers and isolation caps to quickly secure the area and protect your home.
- **Common misunderstanding:** "The flushing machine uses high pressure that physically bursts strong, healthy pipes."
- **Reality:** The machine operates at low pressure (under 2 bar). Leaks occur not because of excess pressure, but because removing old rust deposits exposes pre-existing corrosion holes.
- **Dangerous oversimplification to avoid:** Saying "don't worry, it's completely safe" can leave the installer liable if an old radiator leaks during a flush.
- **Best analogy families:** Body/health.
- **Where analogy works:** Comparing a corroded radiator to a decayed tooth. The sludge is like tartar buildup; when a dentist cleans it away, they aren't creating a cavity—they're exposing pre-existing decay that now needs to be properly addressed.
- **Where analogy breaks:** Teeth do not flood a living room floor when they leak, so the financial and property damage risks are not directly comparable.
- **Best graphic:** A diagram of a rusted radiator wall. The left panel shows a rust hole plugged with black sludge, holding water back. The right panel shows the sludge washed away, leaving a pinhole leak.
- **Best animation:** A cross-section animation showing water carrying away sludge, revealing a thin metal wall that begins to weep water, followed by an engineer turning an isolation valve to stop it.
- **Best print sheet:** A "System Health Disclaimer and Assessment Checklist" that clearly outlines risks, mitigation steps, and pre-flush signature requirements.
- **Portal treatment:** A digital risk disclaimer. Customers with older systems must check boxes acknowledging they understand the risk before confirming a Powerflush.
- **PDF treatment:** A clear "System Age and Structural Integrity Notice" placed directly alongside the flushing options in the quote PDF.
- **Engineer handover treatment:** A mandatory pre-flush survey form to document pre-existing weep spots, rust blisters, or pipe degradation.
- **Technical appendix:** Risk assessment parameters. Systems with paper-thin steel radiators (wall thickness <1.0\text{ mm}) or localized pitting corrosion should be cleaned using chemical dispersion agents over 14 days rather than high-velocity mechanical flushing.
- **Accessibility notes:**
    - *Dyslexia:* Use simple icons for risk and safety.
    - *ADHD:* Bold key phrases to keep readers focused on the main risks.
    - *Low literacy:* Explain the process as "revealing pre-existing wear" rather than "dislodging mechanical debris."
    - *Technophobia:* Clearly outline the physical backup steps the engineer takes.
    - *Visual impairment:* Provide alt-text and high-contrast color choices.
    - *Motion sensitivity:* Use simple static slide transitions rather than smooth zoom-in animations.

- **Priority:** MVP
- **Related concepts:** CON_F01, CON_F07.
- **Suggested diagram ID:** DIA_F03_LEAK_RISK
- **Suggested animation ID:** ANI_F03_PINHOLE_EXPOSURE
- **Suggested print card ID:** CRD_F03_RISK_ASSESS

Focus 4: Magnetic Filters and Sludge

- **Concept ID:** CON_F04
- **Plain-English title:** Your Heating System's Magnetic Shield
- **Category:** Cleaning, Water Quality, and Protection
- **Customer question it answers:** "What is a magnetic filter, and why does my heating system need one permanently installed?"
- **Why it matters:** Even after flushing, iron components inside radiators will continue to corrode over time, creating black magnetite sludge. A magnetic filter captures this circulating sludge before it can reach and damage your boiler or heat pump.
- **Atlas trigger conditions:** Included by default on all boiler and heat pump system designs generated by the Atlas engine.
- **Engine facts needed:** Primary circuit pipe size (typically 22\text{ mm} or 28\text{ mm}), available physical space near the boiler return pipe, and system volume.
- **Customer wording:** "As water flows through your iron radiators, it naturally produces tiny, sharp particles of magnetic rust. A magnetic filter is a small cylinder fitted with an ultra-strong magnet that attracts and traps these metallic particles, keeping your heating water clean and protecting your system."
- **One-line summary:** A magnetic filter is a permanent safety device that uses a powerful magnet to continuously capture metallic rust before it can damage your heating system.
- **What you may notice:** The engineer will clean the filter during your annual boiler service, rinsing away a thick, black clay-like paste of captured rust.
- **What stays familiar:** The filter runs silently in the background, requiring no electricity and no daily interaction from you.
- **What not to worry about:** The filter is completely sealed and designed with a bypass path, meaning it will never block the flow of water even if it is completely full of rust.
- **Common misunderstanding:** "Installing a magnetic filter means I never have to flush my system again."
- **Reality:** A filter is a protective shield, not a deep-cleaning machine. It traps ongoing debris but cannot clear out thick, heavy sludge that has already settled inside your radiators.
- **Dangerous oversimplification to avoid:** Describing the filter as a "water purifier" can confuse customers into thinking it softens water or removes limescale.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing a magnetic filter to a car engine's oil filter. Just as an oil filter catches tiny metal shavings to keep your engine running smoothly, a magnetic filter catches radiator rust to protect your boiler.
- **Where analogy breaks:** Car oil filters are completely replaced during a service, while magnetic heating filters are simply opened, cleaned, and reassembled.
- **Best graphic:** A cross-section diagram of a magnetic filter. Clear water enters, flows past an ultra-strong central magnet coated in captured black sludge, and exits as clean water.
- **Best animation:** A 3D animation of water carrying jagged black rust particles into a

chamber. The particles are instantly pulled to a central magnetic rod, leaving clear, clean water to flow back to the boiler.
- **Best print sheet:** An A5 "Homeowner's Guide to Filter Care," explaining how the device is serviced and how it preserves your boiler warranty.
- **Portal treatment:** A 3D interactive model. Users click and drag to pull out the virtual magnet and see how rust particles gather on its surface.
- **PDF treatment:** A dedicated component callout box in the quotation PDF: "Your Included Protection Shield: The In-Line Magnetic Filter."
- **Engineer handover treatment:** A live demonstration during handover, showing the customer where the filter is located and confirming it has been fitted on the primary return pipe.
- **Technical appendix:** Hydraulic performance data. Installing a high-performance neodymium magnetic filter (e.g., TF1 or MagnaClean) must not introduce a pressure drop exceeding 2\text{ kPa} at a design flow rate of 1.5\text{ m}^3\text{/h}.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the water flow path.
  - *ADHD:* Keep text blocks short and bullet points clear.
  - *Low literacy:* Focus on the terms "rust trap" and "magnetic shield."
  - *Technophobia:* Reassure the user that the filter operates passively and requires no programming.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the filter assembly diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_F02, CON_F07, CON_F08.
- **Suggested diagram ID:** DIA_F04_MAG_FILTER
- **Suggested animation ID:** ANI_F04_MAGNET_CAPTURE
- **Suggested print card ID:** CRD_F04_FILTER_GDE

Focus 5: Boiler Sizing and Why Bigger is Not Better

- **Concept ID:** CON_B01
- **Plain-English title:** Why a Bigger Boiler is Not Always Better
- **Category:** Boiler and Heat-Source Behaviour
- **Customer question it answers:** "My old boiler was 30kW. Why is Atlas recommending a smaller 18kW boiler? Won't my house be cold on winter days?"
- **Why it matters:** Oversized boilers satisfy small heat demands too quickly, causing them to cycle on and off constantly. This "short cycling" wastes fuel, increases wear on components, and prevents the boiler from condensing efficiently.
- **Atlas trigger conditions:** Property heat loss calculation indicates a peak space heating demand that is significantly lower than the existing boiler's capacity.
- **Engine facts needed:** Calculated whole-house peak heat loss (kW), existing boiler capacity (kW), and minimum modulation limit of the recommended boiler.
- **Customer wording:** "An oversized boiler is like an engine that can only run at full speed or completely off. It heats your radiators up too quickly, shuts down, cools down, and restarts a few minutes later. This constant starting and stopping is highly inefficient and wears out components rapidly."
- **One-line summary:** Sizing your boiler to precisely match your home's heat loss allows it

to run smoothly and continuously, saving energy and extending its lifespan.

- **What you may notice:** The boiler runs for longer, quieter periods instead of firing up with a loud burst and shutting down shortly after.
- **What stays familiar:** Your home remains consistently warm, even on the coldest winter days, because the boiler output still matches your peak heat loss.
- **What not to worry about:** A smaller, correctly sized boiler will not struggle to heat your home; it is simply designed to work steadily rather than in inefficient bursts.
- **Common misunderstanding:** "A bigger boiler heats up my house faster."
- **Reality:** Your radiators can only transfer a set amount of heat to a room, regardless of how large the boiler is. An oversized boiler simply heats up the water inside itself too quickly and shuts down before the room can actually warm up. Increasing radiator size is the only way to deliver more warmth.
- **Dangerous oversimplification to avoid:** Recommending a smaller boiler without clarifying the difference between space heating needs and hot water needs. Combi boilers often need to be larger (e.g., 30kW) solely to heat instant hot water, even if the heating need is only 8kW.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing a boiler to a car. If you only drive in stop-and-go city traffic, a massive V8 engine that accelerates rapidly and slams on the brakes every fifty yards is incredibly inefficient. A smaller, steady engine is much smoother and uses far less fuel.
- **Where analogy breaks:** Cars rely on mechanical gears, whereas boilers use fuel modulation valves to scale their output down.
- **Best graphic:** A dual-timeline diagram. The top timeline shows an oversized boiler cycling on/off six times an hour. The bottom timeline shows a correctly sized modulating boiler running in one smooth, continuous block.
- **Best animation:** An animation showing a temperature gauge. Watch an oversized boiler cycle rapidly between blue and red, while a modulating boiler settles smoothly into a steady, green efficiency zone.
- **Best print sheet:** An A5 "Sizing Myths Debunked" card that compares space heating calculations with hot water demands.
- **Portal treatment:** An interactive graph. Users adjust a slider showing boiler size to see how it affects cycling frequency and fuel bills.
- **PDF treatment:** A dedicated section in the proposal explaining the exact heat loss calculation and why the recommended capacity is chosen.
- **Engineer handover treatment:** Software settings verification, ensuring the boiler's maximum space heating output is range-rated to match the calculated heat loss.
- **Technical appendix:** Sizing calculations. Space heating systems must be sized to meet heat loss at local design temperatures, using standard sizing models without arbitrary safety margins, as required by Part L.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual timelines instead of text-heavy tables.
    - *ADHD:* Bold key phrases to keep readers focused on modulation benefits.
    - *Low literacy:* Focus on the terms "smooth running" and "steady warmth."
    - *Technophobia:* Reassure the user that the system manages these cycles automatically.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the timeline diagrams.

    - *Motion sensitivity:* Disable autoplay on the animation, using a play/pause toggle instead.
- **Priority:** MVP
- **Related concepts:** CON_B04, CON_B05, CON_E03.
- **Suggested diagram ID:** DIA_B01_BOILER_SIZING
- **Suggested animation ID:** ANI_B01_MODULATION_CYCLE
- **Suggested print card ID:** CRD_B01_SIZE_MYTHS

Focus 6: Gas Pipe Sizing

- **Concept ID:** CON_B02
- **Plain-English title:** Why Your Gas Pipe May Need an Upgrade
- **Category:** Boiler and Heat-Source Behaviour
- **Customer question it answers:** "My new boiler is going in the exact same spot as my old one. Why do you need to replace the gas pipe leading to it?"
- **Why it matters:** Modern boilers require a high volume of gas to run efficiently and safely, especially combi boilers during hot water use. If the existing pipe is too narrow (15\text{ mm}), the gas pressure will drop under load, causing the boiler to lock out or burn fuel incompletely.
- **Atlas trigger conditions:** A high-power combi boiler (>30\text{ kW}) is being installed, and the existing gas line is 15\text{ mm} for a run longer than 6 meters.
- **Engine facts needed:** Boiler heat input rating (kW), total gas pipe run length (m), fitting counts, and existing pipe diameters.
- **Customer wording:** "Just like a high-power shower needs a wide water pipe to deliver a strong flow, a modern boiler needs a wide gas pipe (22\text{ mm}) to receive enough fuel when running at full capacity. If the pipe is too narrow, the boiler will starve for fuel, lock out, or run inefficiently."
- **One-line summary:** Upgrading your gas pipe ensures your boiler receives a safe, steady flow of fuel without experiencing pressure drops under full load.
- **What you may notice:** The engineer will run a thicker copper pipe (22\text{ mm} or 28\text{ mm}) from your gas meter to the boiler, which may require lifting floorboards along the route.
- **What stays familiar:** The gas meter and your other gas appliances remain in the same locations.
- **What not to worry about:** Lifting floorboards is a routine part of installation. Engineers protect your floors and carefully replace all boards afterward.
- **Common misunderstanding:** "My old boiler worked fine on a 15\text{ mm} pipe, so my new one will too."
- **Reality:** Older boilers were less powerful or had lower instant flow demands. Modern combis heat hot water instantly, requiring up to 35\text{ kW} of gas demand. This requires a wider pipe to prevent pressure drops.
- **Dangerous oversimplification to avoid:** Describing the pipe upgrade as "just a recommendation" when it is a mandatory safety requirement under BS 6891.
- **Best analogy families:** Water/tank/straw.
- **Where analogy works:** Comparing a narrow gas pipe to drinking a thick milkshake through a thin straw. You have to suck incredibly hard (pressure drop) to get very little fluid. Upgrading to a wider straw allows you to drink easily and smoothly.
- **Where analogy breaks:** Milkshake is non-hazardous, whereas gas pressure drops can

lead to carbon monoxide risks from incomplete combustion.
- **Best graphic:** A comparative diagram showing 15\text{ mm} vs. 22\text{ mm} pipe bores with gas molecules crowding in the narrower pipe, resulting in a low flame at the burner.
- **Best animation:** An animation of a boiler starting up. Watch the pressure gauge at the meter drop below safe limits as a narrow gas line starves the burner, followed by a wider line maintaining a steady green pressure zone.
- **Best print sheet:** An A5 "Gas Sizing Guide" explaining pressure drops and outlining the installation process for homeowners.
- **Portal treatment:** An interactive pipe routing map. Users see how the length of the run and the number of bends dictate the required pipe thickness.
- **PDF treatment:** A clear section in the quote booklet justifying the cost and disruption of a gas line upgrade based on safety standards.
- **Engineer handover treatment:** Commissioning checks to verify gas pressure drops stay within the 1\text{ mbar} limit at maximum load, as required by BS 6891.
- **Technical appendix:** Hydraulic design standards. Low-pressure gas installations must limit pressure drops to 1\text{ mbar} between the meter outlet and the appliance inlet, as specified in BS 6891.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the gas pipe diameter.
  - *ADHD:* Keep text blocks short and bullet points clear.
  - *Low literacy:* Focus on the terms "gas safety" and "fuel starvation."
  - *Technophobia:* Reassure the user that the engineer handles these safety checks.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the pipe cross-section diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_B01, CON_I05.
- **Suggested diagram ID:** DIA_B02_GAS_FLOW
- **Suggested animation ID:** ANI_B02_PRESSURE_DROP
- **Suggested print card ID:** CRD_B02_GAS_UPGRADE

Focus 7: System Pressure and Filling Loop

- **Concept ID:** CON_B03
- **Plain-English title:** How to Top Up Your Heating System's Pressure
- **Category:** Boiler and Heat-Source Behaviour
- **Customer question it answers:** "My boiler has stopped working and is showing a 'low pressure' error code. How do I fix this myself?"
- **Why it matters:** Sealed systems require a minimum water pressure (usually 1.0\text{ bar}) to move heat safely. Over time, tiny micro-leaks can cause pressure to drop, triggering a safety shutdown to prevent the system from dry-firing.
- **Atlas trigger conditions:** Customer experiences a low-pressure lockout, or a sealed system is recommended.
- **Engine facts needed:** Recommended system pressure (typically 1.0\text{--}1.5\text{ bar}), location of the filling loop, and low-pressure safety switch parameters.
- **Customer wording:** "Your heating system is a sealed loop of pressurized water. If the pressure drops too low, a safety switch turns the boiler off to protect it. You can easily top it up yourself using the silver-braided filling loop under your boiler."

- **One-line summary:** Sealed heating systems require a steady water pressure of 1.0\text{ bar} to run safely, which you can easily top up using a manual filling loop.
- **What you may notice:** The digital display or round pressure gauge on your boiler will drop below 0.5\text{ bar}, and a warning light or error code will appear.
- **What stays familiar:** Once topped up, your boiler resets and resumes heating your home normally.
- **What not to worry about:** A minor pressure drop once or twice a year is completely normal and caused by tiny amounts of air escaping through automatic vents.
- **Common misunderstanding:** "Low boiler pressure means my gas supply is running low."
- **Reality:** Boiler pressure refers strictly to the water inside your radiator loop, not your gas supply. It is entirely separate from gas pressure.
- **Dangerous oversimplification to avoid:** Telling a customer to "just open the valves" without reminding them to close both valves completely afterward, which can over-pressurize the system and cause the safety valve to leak.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing your heating system's water pressure to your car's tire pressure. If a tire drops below safe limits, a warning light appears on your dashboard, and you need to pump in a little air to return it to the correct pressure.
- **Where analogy breaks:** Car tires are filled with air, whereas a heating system is filled with water.
- **Best graphic:** A step-by-step diagram showing the filling loop. Panel 1: Low pressure gauge (0.2\text{ bar}). Panel 2: Open first valve, then slowly open the second. Panel 3: Watch the gauge rise to 1.0\text{ bar} and close both valves.
- **Best animation:** A short, high-contrast video tutorial. Watch a hand turn the filling loop valves, showing the pressure needle rise into the green zone in real time, followed by the valves being closed.
- **Best print sheet:** An A5 laminated card designed to be hung directly on the filling loop, featuring simple, icon-led instructions.
- **Portal treatment:** A guided interactive troubleshooter. Users answer questions about their boiler's pressure reading to get tailored steps or book a service visit.
- **PDF treatment:** A dedicated "Maintenance Quick Guide" page included in the digital handover pack.
- **Engineer handover treatment:** A hands-on tutorial where the engineer physically demonstrates how to use the filling loop during handover.
- **Technical appendix:** System pressure limits. The pressure relief valve (PRV) is factory-set to open at 3.0\text{ bar}. The low-pressure safety switch typically cuts power to the burner at 0.5\text{[span_460](start_span)[span_460](end_span) bar}.
- **Accessibility notes:**
  - *Dyslexia:* Use high-contrast visual status indicators (Green/Red) on the filling loop instructions.
  - *ADHD:* Keep step-by-step instructions short and numbered.
  - *Low literacy:* Focus on the terms "topping up water" and "pressure gauge."
  - *Technophobia:* Provide clear video examples with simple, slow movements.
  - *Visual impairment:* Ensure the gauge has large, high-contrast numbers.
  - *Motion sensitivity:* Use simple static slide transitions rather than smooth zoom-in animations.
- **Priority:** MVP

- **Related concepts:** CON_A01, CON_B01, CON_B11.
- **Suggested diagram ID:** DIA_B03_FILLING_LOOP
- **Suggested animation ID:** ANI_B03_PRESSURE_REFILL
- **Suggested print card ID:** CRD_B03_REPRESSURIZE

Focus 8: Open-Vented to Sealed Conversion

- **Concept ID:** CON_A01
- **Plain-English title:** Converting Your Heating to a Modern Sealed Loop
- **Category:** System Identity and Topology
- **Customer question it answers:** "My quote recommends converting my old heating system to a 'sealed system.' What does this mean, and what are the benefits?"
- **Why it matters:** Legacy systems rely on open tanks in the loft to handle water expansion, which continuously introduces fresh oxygen, accelerating internal corrosion. Converting to a sealed system removes these tanks, reduces corrosion, improves water circulation, and frees up loft space.
- **Atlas trigger conditions:** Property has a regular boiler with an open F&E tank in the loft, and is transitioning to a modern system boiler, combi boiler, or heat pump.
- **Engine facts needed:** Presence of loft tanks, age of pipework, radiator condition, and calculated system volume.
- **Customer wording:** "Older systems have open water tanks in the loft that can freeze, leak, or introduce fresh air that rusts your radiators from the inside out. Converting to a sealed system removes these tanks entirely, replacing them with a compact, pressurized vessel that sits inside your home, creating a clean, efficient, and leak-free loop."
- **One-line summary:** Converting to a sealed system replaces drafty loft tanks with a compact, pressurized expansion vessel inside your home to reduce corrosion and free up space.
- **What you may notice:** The engineer will remove the dusty water tanks from your loft, install a compact expansion vessel (often inside the boiler), and pressurize your system.
- **What stays familiar:** Your radiators and pipework layouts remain the same, but they will heat up faster and run more quietly.
- **What not to worry about:** Removing loft tanks is a routine process. Any old pipes leading to the loft are safely capped and insulated to prevent drafts.
- **Common misunderstanding:** "Converting to a pressurized system will cause my old radiators to burst."
- **Reality:** While healthy radiators handle the pressure easily, converting does increase system pressure from around 0.2\text{ bar} to 1.0\text{ bar}. A professional pre-install survey is essential to identify and replace any fragile radiators before pressurizing.
- **Dangerous oversimplification to avoid:** Failing to mention that old, buried pipework (such as pipes in concrete floor screeds) must be pressure-tested first to ensure it can handle the conversion.
- **Best analogy families:** Factual/no analogy.
- **Where analogy works:** Comparing the conversion to sealing a bottle of soda. An open bottle lets air in, causing the soda to go flat and spill easily if tipped. A sealed bottle stays clean, pressurized, and completely self-contained.
- **Where analogy breaks:** Soda bottles do not heat a home, so the analogy fails to capture thermal expansion dynamics.
- **Best graphic:** A comparative diagram. The left panel shows an open-vented system with

a loft tank and cold-feed pipes. The right panel shows a sealed system with the tanks removed and a pressurized expansion vessel installed.
- **Best animation:** A 2D animation showing water heating up and expanding. Watch it rise into a cold loft tank in an open system, then watch it expand safely against a rubber bladder inside a compact expansion vessel in a sealed system.
- **Best print sheet:** An A5 "Sealing Your System" guide detailing the conversion process, benefits, and pre-install checks.
- **Portal treatment:** An interactive home graphic. Users click on the loft to see the tanks disappear and click on the boiler to see the new expansion vessel in place.
- **PDF treatment:** A dedicated page in the proposal: "System Modernization: Removing Loft Tanks for Efficiency."
- **Engineer handover treatment:** A visual check of the capped loft pipes and a tutorial showing the customer the new pressure gauge.
- **Technical appendix:** Structural details. Open-vented systems provide a static head of around 0.1\text{--}0.3\text{ bar}. Sealed conversions operate at a cold fill pressure of 1.0\text{--}1.2\text{ bar}. Installers must pressure-test existing pipework to 1.5\text{ bar} before commissioning.
- **Accessibility notes:**
  - *Dyslexia:* Use high-contrast visual timelines instead of text-heavy tables.
  - *ADHD:* Keep step-by-step conversion steps short and numbered.
  - *Low literacy:* Explain the process as "removing loft tanks" rather than "decoupling the open vent."
  - *Technophobia:* Reassure the user that the system runs automatically once sealed.
  - *Visual impairment:* Ensure all image-heavy pages are readable by screen-reading software.
  - *Motion sensitivity:* Disable autoplay on the animation, using a play/pause toggle instead.
- **Priority:** MVP
- **Related concepts:** CON_A05, CON_B03, CON_F07.
- **Suggested diagram ID:** DIA_A01_SEALED_CONV
- **Suggested animation ID:** ANI_A01_EXPANSION_VESSEL
- **Suggested print card ID:** CRD_A01_SEALED_SYS

Focus 9: Unvented Cylinder Safety

- **Concept ID:** CON_C01
- **Plain-English title:** How Your Hot Water Cylinder Keeps Itself Safe
- **Category:** Hot Water Systems
- **Customer question it answers:** "I've heard unvented hot water cylinders can be dangerous if they overheat. Is this true, and how do they keep my home safe?"
- **Why it matters:** Because unvented cylinders connect directly to the water mains, they run under pressure. To prevent over-pressurization and thermal expansion risks, they require built-in safety devices, G3-qualified installers, and clear discharge pathways.
- **Atlas trigger conditions:** An unvented cylinder installation is recommended.
- **Engine facts needed:** Cylinder storage volume (L), mains pressure, and G3 certification status of the installation team.
- **Customer wording:** "Unvented cylinders are incredibly robust, heavy-duty steel tanks designed to handle high water pressure safely. They feature multiple layers of safety

protection, including automatic thermostats, pressure valves, and a visible overflow funnel (tundish) to ensure your hot water never exceeds safe limits."

- **One-line summary:** Unvented cylinders are highly secure, mains-pressure tanks protected by three layers of safety devices to ensure safe, stable hot water.
- **What you may notice:** A silver-braided expansion vessel near the cylinder, several safety brass valves, and a small plastic funnel (tundish) on the pipework with a clear window.
- **What stays familiar:** You enjoy powerful, mains-pressure hot water at all your taps and showers simultaneously.
- **What not to worry about:** Seeing a tiny drip inside the tundish once or twice a year is completely normal and simply means a valve is releasing a tiny bit of expansion pressure.
- **Common misunderstanding:** "Unvented cylinders are prone to exploding like bombs."
- **Reality:** While unsealed, unregulated boilers from a century ago had risks, modern unvented cylinders feature triple-redundant mechanical safety devices. They are exceptionally safe and must be installed by certified G3 experts by law.
- **Dangerous oversimplification to avoid:** Describing the tundish as "just a plastic pipe" without explaining its role as a visible air break to prevent dirty drain water from backing up into your cylinder.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing unvented cylinder safety to your car's brakes. You have a main foot brake, an emergency handbrake, and built-in electronic ABS systems. If one system ever faces an issue, the others instantly step in to keep you safe.
- **Where analogy breaks:** Brakes are active components, whereas cylinder safeties are passive relief valves.
- **Best graphic:** A detailed diagram showing the cylinder, the expansion vessel, the temperature relief valve, the pressure relief valve, and the tundish discharge pathway.
- **Best animation:** A 3D animation showing water heating and expanding. Watch the excess pressure expand into the red expansion vessel, followed by a simulated overheat where the relief valve opens, letting water flow safely down the tundish.
- **Best print sheet:** An A5 "Cylinder Safety and Tundish Guide" explaining what the safety devices do and how to inspect them.
- **Portal treatment:** An interactive safety tour. Users hover over components (such as the TPRV and tundish) to read simple explanations of how they work.
- **PDF treatment:** A dedicated section in the proposal emphasizing compliance with G3 Building Regulations and verifying the team's professional certifications.
- **Engineer handover treatment:** A physical demonstration of the safety valves and a check to ensure the tundish is clear and visible.
- **Technical appendix:** Regulatory compliance rules. Under Building Regulations G3, systems over 15 liters must discharge safely. The D1 pipe must be metal, under 600\text{ mm}, and run vertically. The D2 pipe must be at least one size larger and maintain a continuous downward fall of 1:200.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the water flow path.
    - *ADHD:* Keep step-by-step safety checks short and numbered.
    - *Low literacy:* Focus on the terms "safety valves" and "overflow funnel."
    - *Technophobia:* Reassure the user that the system has passive mechanical safeties.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the cylinder assembly diagrams.

    - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_A05, CON_C04, CON_D01.
- **Suggested diagram ID:** DIA_C01_CYLINDER_G3
- **Suggested animation ID:** ANI_C01_TUNDISH_FLOW
- **Suggested print card ID:** CRD_C01_SAFETY_CHECK

Focus 10: Pressure vs. Storage

- **Concept ID:** CON_C02
- **Plain-English title:** Hot Water: Instant Flow vs. Reliable Storage
- **Category:** Hot Water Systems
- **Customer question it answers:** "Should I get a combi boiler that heats water instantly, or a system boiler with a hot water storage tank?"
- **Why it matters:** Selecting the incorrect hot water system can lead to disappointing shower pressure, long bath fill times, or high energy use if sized incorrectly.
- **Atlas trigger conditions:** Property is undergoing a system redesign, and the heating engine is evaluating combi boilers vs. stored hot water cylinders.
- **Engine facts needed:** Household occupancy, total bathroom count, simultaneous shower demand, and incoming cold mains flow rate.
- **Customer wording:** "A combi boiler heats water instantly on demand, which is perfect for smaller homes with one bathroom. A stored hot water cylinder keeps a large volume of water pre-heated at mains pressure, allowing you to run multiple high-power showers at the same time without any drop in performance."
- **One-line summary:** Combi boilers are ideal for small, single-bathroom homes, while stored hot water cylinders are essential for large families with high, simultaneous demands.
- **What you may notice:** With a stored cylinder, your baths fill rapidly and your showers stay powerful, even if someone else turns on a tap downstairs.
- **What stays familiar:** Hot water flows instantly from your taps, but you may need to wait for a cylinder to reheat if you empty it completely.
- **What not to worry about:** Modern cylinders are highly insulated, meaning they lose very little heat and are incredibly efficient to keep warm.
- **Common misunderstanding:** "A combi boiler is always cheaper and more efficient than a system with a hot water cylinder."
- **Reality:** While combis do not have standing heat losses, they must fire at full capacity to heat water instantly, which can be less efficient. For larger homes with high hot water demands, a stored cylinder is far more efficient and reliable.
- **Dangerous oversimplification to avoid:** Recommending a combi boiler to a large family solely to save cupboard space, which can lead to frustrating temperature drops when multiple taps are used at once.
- **Best analogy families:** Kitchen/cooking.
- **Where analogy works:** Comparing a combi boiler to an instant electric kettle that heats water as you pour it, and a stored cylinder to a large pre-filled thermal flask. The flask can fill ten cups of tea instantly, while the kettle can only boil one cup at a time.
- **Where analogy breaks:** Kettles operate on electricity, whereas domestic hot water systems usually rely on gas or heat pump refrigerant loops.
- **Best graphic:** A split comparative diagram. The left side shows a combi boiler struggling

to feed a second shower. The right side shows a stored cylinder feeding two powerful showers simultaneously.
- **Best animation:** An animation showing water flow. Watch a single shower drop to a trickle when a second tap is opened in a combi system, then watch both outlets maintain full, powerful flow in a cylinder system.
- **Best print sheet:** An A5 "Hot Water Sizing Guide" comparing flow rates, fill times, and suitability for different home sizes.
- **Portal treatment:** An interactive "Home Planner." Users select their bathroom count and family size to see which system type matches their lifestyle.
- **PDF treatment:** A dedicated comparison page in the quote booklet justifying the recommended system choice based on daily usage.
- **Engineer handover treatment:** Sizing verification checks to ensure the cylinder recovery rate matches the boiler's output.
- **Technical appendix:** System design criteria. Combi boilers are limited by their instant heat transfer capacity (typically 24\text{--}35\text{ kW} for a 10\text{--}15\text{ L/min} flow at a 35\text{ K} rise). Cylinders can deliver mains flow rates exceeding 40\text{ L/min} simultaneously, limited only by the cold water main.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the hot water flow path.
  - *ADHD:* Keep step-by-step system comparisons short and numbered.
  - *Low literacy:* Focus on the terms "instant heat" and "stored hot water."
  - *Technophobia:* Reassure the user that the system runs automatically once set.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_A03, CON_C04, CON_D01.
- **Suggested diagram ID:** DIA_C02_FLOW_COMP
- **Suggested animation ID:** ANI_C02_PRESSURE_DEMAND
- **Suggested print card ID:** CRD_C02_HOT_WATER_GDE

Focus 11: Static Pressure vs. Dynamic Flow

- **Concept ID:** CON_D01
- **Plain-English title:** Static Water Pressure vs. Dynamic Flow
- **Category:** Water Supply and Hydraulics
- **Customer question it answers:** "My bathroom pressure seems great when I turn a single tap on, but it drops to a trickle when I run a second tap. Why is this happening?"
- **Why it matters:** Relying on simple pressure measurements without checking flow rate can lead to installing systems that underperform when multiple taps are used at once.
- **Atlas trigger conditions:** Property is undergoing a technical survey, and the engine is evaluating system suitability for high-demand hot water.
- **Engine facts needed:** Static pressure (bar), dynamic flow rate (L/min), and incoming water main pipe material.
- **Customer wording:** "Static pressure is how hard water pushes when all your taps are closed. Dynamic flow is the actual volume of water your pipes can deliver when taps are open. If your incoming water main is narrow or shared, the pressure will drop as soon as you open a second tap, reducing the flow to a trickle."

- **One-line summary:** Static pressure is the force of still water, while dynamic flow is the actual volume of water delivered when taps are running.
- **What you may notice:** The surveyor will attach a pressure gauge to an outside tap, then open other taps in the house to watch the pressure drop.
- **What stays familiar:** The static pressure reading remains high when everything is off, but your dynamic flow rate is what determines your actual shower performance.
- **What not to worry about:** If your dynamic flow rate is low, there are excellent modern solutions (such as accumulators or booster pumps) to fix it.
- **Common misunderstanding:** "I have high water pressure, so my system can easily run three showers at once."
- **Reality:** High static pressure simply means the water is pushing hard when still. If your incoming pipe is narrow, friction will cause that pressure to drop as soon as you open a tap, restricting the flow of water.
- **Dangerous oversimplification to avoid:** Recommending a high-flow unvented cylinder based solely on a high static pressure reading, which can result in disappointing performance if the dynamic flow is restricted.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing static pressure to a speed limit (how fast you could go on an empty road) and dynamic flow to actual traffic volume (how many cars actually get through during rush hour). A high speed limit doesn't help if a narrow lane creates a traffic jam.
- **Where analogy breaks:** Traffic involves driver behavior, whereas water flow is governed purely by hydraulic friction and pressure.
- **Best graphic:** A comparative diagram. The left panel shows a closed tap with a high pressure reading (3.0\text{ bar}). The right panel shows two open taps with the pressure needle dropping to 0.5\text{ bar} due to a narrow pipe.
- **Best animation:** An animation of water moving through a pipe. Watch water flow easily through a wide pipe when multiple taps are opened, then watch water slow down and back up when squeezed through a narrow pipe.
- **Best print sheet:** An A5 "Pressure vs. Flow Guide" explaining why both measurements are critical for shower performance.
- **Portal treatment:** An interactive diagnostic card. Users enter their tap and shower count to see how dynamic flow rates affect daily performance.
- **PDF treatment:** A dedicated technical breakdown in the survey report explaining the incoming water main measurements.
- **Engineer handover treatment:** Flow rate commissioning checks to ensure the system delivers the specified dynamic flow rates.
- **Technical appendix:** Hydraulic performance limits. Static pressure must be measured using a calibrated gauge. Dynamic flow must be verified using a weir gauge or flow cup, ensuring it does not drop below 15\text{ L/min} at 1.5\text{ bar} residual pressure.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the water flow path.
    - *ADHD:* Keep step-by-step system comparisons short and numbered.
    - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
    - *Technophobia:* Reassure the user that the system runs automatically once set.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
    - *Motion sensitivity:* Use simple static animations with clear labels.

- **Priority:** MVP
- **Related concepts:** CON_C04, CON_D02, CON_D05.
- **Suggested diagram ID:** DIA_D01_PRESSURE_FLOW
- **Suggested animation ID:** ANI_D01_DYNAMIC_FLOW
- **Suggested print card ID:** CRD_D01_FLOW_GUIDE

Focus 12: Accumulators vs. Booster Pumps

- **Concept ID:** CON_D02
- **Plain-English title:** Boosting Your Water Flow: Quiet Accumulators vs. Powered Pumps
- **Category:** Water Supply and Hydraulics
- **Customer question it answers:** "My mains water pressure is low. Should I install a cold water accumulator or a powered booster pump to fix it?"
- **Why it matters:** Selecting the incorrect water pressure solution can lead to annoying pump noise, high energy bills, or systems that fail to deliver enough water during peak times.
- **Atlas trigger conditions:** Property is undergoing a technical survey, and the incoming mains water flow rate is under 12\text{ L/min}.
- **Engine facts needed:** Incoming mains flow rate (L/min), static pressure, occupancy, and available installation space.
- **Customer wording:** "An accumulator is a pressurized vessel that stores a large volume of water under your home's natural pressure, delivering a quiet, powerful boost when you need it. A powered booster pump uses electricity to actively pump water from a storage tank, making it ideal for very large homes with high, continuous hot water demands."
- **One-line summary:** Accumulators provide a quiet, energy-free pressure boost by storing water, while powered pumps actively push water to meet very high demands.
- **What you may notice:** With an accumulator, your water runs powerfully and silently. With a booster pump, you may hear a low hum when running taps.
- **What stays familiar:** You enjoy powerful, mains-pressure showers throughout your home.
- **What not to worry about:** Both systems are highly reliable and designed to keep your showers powerful, even during busy mornings.
- **Common misunderstanding:** "Accumulators use electricity to actively pump water."
- **Reality:** Accumulators do not require any electricity. They work by using the natural energy of your water mains to compress air inside a sealed bladder, releasing that stored pressure silently when you open a tap.
- **Dangerous oversimplification to avoid:** Recommending a booster pump without checking the local Water Regulations, which limit pumps connected directly to the mains to a maximum flow of 12\text{ L/min}[span_605](start_span)[span_605](end_span)[span_626](start_span)[span_626](end_span). Larger demands require a break tank.
- **Best analogy families:** Water/tank/straw.
- **Where analogy works:** Comparing an accumulator to a stretched rubber balloon. As mains water fills the balloon, it expands and stores energy. When you open a tap, the balloon naturally contracts, pushing the water out powerfully and silently.
- **Where analogy breaks:** Balloons can pop easily, whereas accumulators are heavy-duty steel pressure vessels designed to last for decades.
- **Best graphic:** A comparative diagram. The left side shows an accumulator storing water

silently under pressure. The right side shows a booster pump connected to a storage tank, with electrical connections.
- **Best animation:** An animation showing water flow. Watch an accumulator fill slowly during low-demand hours, then release a powerful boost of water when three taps are opened simultaneously.
- **Best print sheet:** An A5 "Pressure Solutions Guide" comparing costs, space requirements, noise levels, and electricity use for both systems.
- **Portal treatment:** An interactive home graphic. Users click on different pressure solutions to see where they are installed and how they affect daily shower performance.
- **PDF treatment:** A dedicated section in the proposal explaining the dynamic flow calculations and justifying the recommended solution.
- **Engineer handover treatment: Commissioning checks to verify pressure and flow rates meet the specified design targets.
- **Technical appendix:** System design parameters. Accumulators operate based on Boyle's Law (P_1 V_1 = P_2 V_2) and typically deliver around 60\% of their total volume as usable water. Pumps connected directly to the mains must not exceed a flow rate of 12\text{ L/min} under Water Regulation 4.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the water flow path.
  - *ADHD:* Keep step-by-step system comparisons short and numbered.
  - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
  - *Technophobia:* Reassure the user that the system runs automatically once set.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_C04, CON_D01, CON_D05.
- **Suggested diagram ID:** DIA_D02_PRESSURE_BOOST
- **Suggested animation ID:** ANI_D02_ACCUMULATOR_FLOW
- **Suggested print card ID:** CRD_D02_BOOST_COMP

Focus 13: Microbore Pipe Limitations

- **Concept ID:** CON_D03
- **Plain-English title:** Understanding Thin (Microbore) Heating Pipes
- **Category:** Water Supply and Hydraulics
- **Customer question it answers:** "My home has very thin copper pipes (8\text{ mm} or 10\text{ mm}). Can I still install a modern heat pump without replacing them?"
- **Why it matters:** Modern low-temperature heat pumps require higher water flow rates than older high-temperature boilers to deliver the same amount of warmth. Thin microbore pipes restrict this flow, which can cause lukewarm radiators, system noise, or efficiency losses if not addressed.
- **Atlas trigger conditions:** Property is undergoing a heat pump survey, and the existing pipework features 8\text{ mm} or 10\text{ mm} copper microbore runs.
- **Engine facts needed:** Pipe diameters, calculated room-by-room heat loss, radiator sizes, and heat pump flow rates.
- **Customer wording:** "Thin microbore pipes (8\text{ mm} or 10\text{ mm}) were popular in older homes because they were easy to bend and install. However, modern heat pumps

run at lower water temperatures, meaning they need to move a larger volume of water to keep your rooms warm. If your pipes are too thin, they can restrict this flow, making radiators furthest from the boiler slow to warm up."

- **One-line summary:** Thin microbore pipes can restrict the higher water flow rates required by modern heat pumps, sometimes requiring simple system adjustments.
- **What you may notice:** The surveyor will measure the thickness of the pipes leading to your radiators and calculate if they can handle the required flow.
- **What stays familiar:** Most of your existing pipework can often be kept, with only the most restricted sections needing an upgrade.
- **What not to worry about:** If keeping your microbore pipes is the best option, engineers can use clever hydraulic separation (like low-loss headers or buffer tanks) to keep your system running smoothly and quietly.
- **Common misunderstanding:** "I have microbore pipes, so I absolutely cannot get a heat pump."
- **Reality:** Heat pumps can run successfully with microbore pipes if the system is designed carefully. This may involve adjusting temperature settings, upgrading specific radiators, or adding a small buffer tank to manage the flow.
- **Dangerous oversimplification to avoid:** Saying "microbore is fine" without checking the length of the runs, which can lead to noisy radiators or system shutdowns in cold weather.
- **Best analogy families:** Water/tank/straw.
- **Where analogy works:** Comparing a microbore pipe to a thin drinking straw. If you need to drink a glass of water quickly (high flow), sucking through a thin straw is hard work. Upgrading to a wider straw allows you to drink easily and comfortably.
- **Where analogy breaks:** Straws carry cold fluids, whereas heating pipes transport hot water under thermal expansion.
- **Best graphic:** A diagram comparing 22\text{ mm}, 15\text{ mm}, and 10\text{ mm} pipe bores with arrows showing water flow rates and friction resistance.
- **Best animation: An animation showing water flow. Watch water struggle and create noise as it is squeezed through a thin microbore pipe, then watch it flow smoothly and quietly through a standard 15\text{ mm} pipe.
- **Best print sheet:** An A5 "Microbore Pipe Guide" explaining how thin pipes affect flow rates and what solutions are available.
- **Portal treatment:** An interactive diagnostic card. Users enter their pipe thickness to see how it affects heat pump compatibility and efficiency.
- **PDF treatment:** A dedicated section in the proposal explaining the hydraulic calculations and justifying any recommended pipe upgrades.
- **Engineer handover treatment:** Commissioning checks to verify flow rates and pressure drops stay within safe limits.
- **Technical appendix:** Hydraulic performance limits. Flow velocity in microbore systems should generally remain under 1.0\text{ m/s} to prevent noise and erosion. Pressure drops should not exceed 350\text{ Pa/m}.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the pipe diameters.
    - *ADHD:* Keep step-by-step system comparisons short and numbered.
    - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
    - *Technophobia:* Reassure the user that the system runs automatically once set.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.

    - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_A02, CON_D04, CON_E02.
- **Suggested diagram ID:** DIA_D03_MICROBORE
- **Suggested animation ID:** ANI_D03_FLOW_RESTRICTION
- **Suggested print card ID:** CRD_D03_MICRO_GUIDE

Focus 14: Hydraulic Balancing

- **Concept ID:** CON_D04
- **Plain-English title:** Balancing Your Heating: Ensuring Every Room Gets Warm
- **Category:** Water Supply and Hydraulics
- **Customer question it answers:** "Why are some rooms in my house warm while others stay cold? How does balancing fix this?"
- **Why it matters:** Water naturally takes the path of least resistance. Without proper balancing, radiators closest to the boiler will steal all the hot water, leaving furthest rooms cold and forcing the system to run inefficiently.
- **Atlas trigger conditions:** Property is undergoing a boiler or heat pump replacement, or has a history of uneven heating.
- **Engine facts needed:** Radiator outputs, pipe lengths, and lockshield valve settings.
- **Customer wording:** "Water is lazy and always takes the easiest shortcut. If your system is unbalanced, hot water will flow straight through the nearest radiators and loop back to the boiler, leaving your furthest radiators cold. Balancing restricts the flow to nearest radiators slightly, pushing hot water evenly to every room."
- **One-line summary:** Hydraulic balancing distributes hot water evenly across all your radiators, ensuring every room reaches its comfortable temperature.
- **What you may notice:** The engineer will attach temperature sensors to your radiator pipes and adjust the small valves (lockshields) on the side.
- **What stays familiar:** Your radiators and thermostat settings remain the same, but every room heats up evenly.
- **What not to worry about:** Adjusting lockshields is a routine part of installation and commissioning, and does not require ongoing adjustments from you.
- **Common misunderstanding:** "Opening all radiator valves fully is the best way to get the most heat."
- **Reality:** Opening all valves fully allows the water to take shortcuts. Furthest radiators will stay cold because water will loop back before reaching them. Restricting nearest valves slightly is what pushes heat to the furthest rooms.
- **Dangerous oversimplification to avoid:** Describing balancing as "just turning knobs" without explaining that it requires precise temperature calculations to ensure the system runs efficiently.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing your heating system to a municipal water supply. If nearest houses have open fire hydrants, water pressure drops, and furthest houses get no water. Restricting nearest flows slightly ensures everyone gets a steady, reliable supply.
- **Where analogy breaks:** Fire hydrants are manual, whereas modern heating valves can adjust dynamically.
- **Best graphic:** A comparative diagram. The left side shows an unbalanced system with

hot water looping through the nearest radiator and leaving the furthest one cold. The right side shows a balanced system with even, warm flow to all radiators.
- **Best animation:** An animation of water flow. Watch hot water take the shortest path in an unbalanced system, then watch the nearest lockshield close slightly, pushing hot water smoothly to the furthest radiator.
- **Best print sheet:** An A5 "Guide to Radiator Balancing" explaining what lockshields do and why they are adjusted.
- **Portal treatment:** An interactive home graphic. Users click on cold rooms to see how hydraulic balancing redistributes heat to keep them warm.
- **PDF treatment:** A dedicated page in the handover pack confirming the system has been fully balanced.
- **Engineer handover treatment:** Verification checks to ensure the temperature difference (\Delta T) across all radiators is balanced and consistent.
- **Technical appendix:** Commissioning rules. Hydronic systems must be balanced to achieve the design temperature differential (\Delta T) across all emitters (typically 11\text{ K} or 20\text{ K} for boilers, and 5\text{--}10\text{ K} for heat pumps), as required by Part L.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the water flow path.
  - *ADHD:* Keep step-by-step system comparisons short and numbered.
  - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
  - *Technophobia:* Reassure the user that the system runs automatically once set.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_D03, CON_E02, CON_I03.
- **Suggested diagram ID:** DIA_D04_BALANCING
- **Suggested animation ID:** ANI_D04_FLOW_BALANCE
- **Suggested print card ID:** CRD_D04_BALANCE_GDE

Focus 15: Low-Loss Headers and Buffers

- **Concept ID:** CON_A02
- **Plain-English title:** Hydraulic Separators: Keeping Your Heat Source Running Smoothly
- **Category:** System Identity and Topology
- **Customer question it answers:** "My quote includes a 'low-loss header' or a 'buffer vessel.' What do these components do, and why does my system need them?"
- **Why it matters:** Boilers and heat pumps require steady water flow to run efficiently, but your radiators are constantly opening and closing. A low-loss header or buffer vessel separates these two loops, ensuring the heat source always has the required flow without being affected by radiator adjustments.
- **Atlas trigger conditions:** Property features multiple heating zones, high-resistance pipework (such as microbore), or a low-temperature heat pump.
- **Engine facts needed:** System volume (L), total zone count, and heat source minimum flow rate requirements.
- **Customer wording:** "A low-loss header is a compact hydraulic separator that acts like a neutral junction box. It separates the water loop inside your boiler or heat pump from the

loop inside your radiators, allowing both to run at their ideal speeds without interfering with one another."

- **One-line summary:** Hydraulic separators decouple your heat source from your radiator loop, ensuring a safe, steady flow of heat and protecting your system.
- **What you may notice:** A compact steel cylinder or insulated tank installed near your boiler or heat pump in the plant room.
- **What stays familiar:** Your radiators heat up normally, but the heat source runs more quietly and efficiently.
- **What not to worry about:** These components run completely silently, require no electricity, and have no moving parts to wear out.
- **Common misunderstanding:** "Low-loss headers and buffer vessels are the exact same component."
- **Reality:** While both provide hydraulic separation, a low-loss header is a compact pipe designed strictly to balance flow rates. A buffer vessel is a larger insulated tank that adds volume to the system, helping to prevent heat pumps from cycling on and off too frequently.
- **Dangerous oversimplification to avoid:** Describing these components as "unnecessary extras" when they are critical for protecting modern heat sources from low-flow damage.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing hydraulic separation to a car's clutch. The clutch decouples the spinning engine from the wheels, allowing the engine to idle smoothly even when the car is stopped. Hydraulic separators allow the heat source to run smoothly even when radiators are closed.
- **Where analogy breaks:** Clutches wear out through friction, whereas hydraulic separators are simple steel chambers with no moving parts.
- **Best graphic:** A diagram showing the primary heat source loop on the left, the secondary radiator loop on the right, and a low-loss header or buffer vessel balancing the flow in the middle.
- **Best animation:** An animation of water flow. Watch radiator valves close, restricting water in the heating loop, while a low-loss header allows water inside the boiler to keep flowing smoothly and safely.
- **Best print sheet:** An A5 "Hydraulic Separation Guide" explaining the difference between low-loss headers, buffer vessels, and volumizers.
- **Portal treatment:** An interactive system map. Users click on different loops to see how hydraulic separation keeps flow rates balanced and efficient.
- **PDF treatment:** A dedicated technical breakdown in the proposal explaining the system design choice.
- **Engineer handover treatment:** Commissioning checks to verify flow rates and temperatures on both sides of the separator.
- **Technical appendix:** Hydraulic separation standards. Decoupling must be designed to ensure the primary circuit pump can maintain the minimum flow rate required by the manufacturer under all operating conditions.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the water flow path.
    - *ADHD:* Keep step-by-step system comparisons short and numbered.
    - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
    - *Technophobia:* Reassure the user that the system runs automatically once set.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the

comparative diagrams.
    - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_A11, CON_D03, CON_H04.
- **Suggested diagram ID:** DIA_A02_HYDRAULIC_SEP
- **Suggested animation ID:** ANI_A02_DECOUPLED_FLOW
- **Suggested print card ID:** CRD_A02_SEP_GUIDE

Focus 16: Radiator Surface Area

- **Concept ID:** CON_E01
- **Plain-English title:** Why Radiator Size Matters for Clean Energy
- **Category:** Radiators, Emitters, and Heat Loss
- **Customer question it answers:** "Why does Atlas recommend upgrading some of my radiators? Can't you just install a larger boiler or heat pump instead?"
- **Why it matters:** Modern low-temperature systems require larger radiator surface areas to deliver the same amount of heat to a room. Upgrading radiators allows the system to run at lower, highly efficient flow temperatures, saving energy and lowering bills.
- **Atlas trigger conditions:** Property is transitioning to a low-temperature system (like a heat pump), and existing radiators are undersized for the calculated heat loss.
- **Engine facts needed:** Calculated room-by-room heat loss, existing radiator surface areas, and design flow temperatures.
- **Customer wording:** "To keep a room warm with cooler water, you need a larger radiator surface area to distribute that heat evenly. Upgrading to a slightly larger or deeper radiator allows your system to run at lower, highly efficient temperatures, saving you money on bills while keeping your home cozy."
- **One-line summary:** Upgrading radiator surface areas allows your heating system to run at lower, highly efficient temperatures while keeping your home perfectly warm.
- **What you may notice:** The engineer will replace some of your radiators with deeper models that feature more convector fins inside.
- **What stays familiar:** Your radiators remain in the same locations, but they heat your home with a gentle, steady warmth.
- **What not to worry about:** Deeper radiators (like Type 22 or 33) fit onto your existing walls easily, often without taking up any extra wall width.
- **Common misunderstanding:** "A larger boiler is the best way to heat up a cold room."
- **Reality:** Your radiators are the "valves" that actually transfer heat into your rooms. If a radiator is too small, a larger boiler will simply overheat the water inside itself and shut down, leaving the room cold. Increasing radiator size is the only way to deliver more warmth.
- **Dangerous oversimplification to avoid:** Recommending larger radiators without verifying that the existing pipework can handle the higher water flow rates required.
- **Best analogy families:** Weather/clothing.
- **Where analogy works:** Comparing radiators to winter blankets. If it gets colder, you don't need a larger heater in the room; you simply need a larger, thicker blanket to keep yourself warm. Upgrading your radiators is like wrapping your home in a cozy, heavy-duty blanket.
- **Where analogy breaks:** Blankets store body heat, whereas radiators actively transfer heat from water into a room.

- **Best graphic:** A diagram comparing radiator types. Show Type 11 (one panel, one set of fins), Type 21 (two panels, one set of fins), and Type 22 (two panels, two sets of fins) to show how surface area can increase without taking up more wall width.
- **Best animation:** An animation showing heat transfer. Watch hot water flow through a small radiator, transferring very little heat, then watch cooler water flow through a larger, deeper radiator, distributing a large, even blanket of heat across the room.
- **Best print sheet:** An A5 "Radiator Selection Guide" showing sizes, depths, and thermal outputs for different radiator types.
- **Portal treatment:** An interactive home graphic. Users click on different rooms to see how upgrading radiators improves comfort and system efficiency.
- **PDF treatment:** A dedicated page in the proposal justifying the recommended radiator upgrades based on heat loss calculations.
- **Engineer handover treatment:** Commissioning checks to verify radiator outputs and flow rates meet the specified design targets.
- **Technical appendix:** Thermal performance limits. Emitter outputs must be calculated using the BS EN 442 standard, ensuring they match room-by-room heat loss at the design flow temperature, as required by Part L.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the water flow path.
  - *ADHD:* Keep step-by-step system comparisons short and numbered.
  - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
  - *Technophobia:* Reassure the user that the system runs automatically once set.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_E02, CON_E03, CON_H02.
- **Suggested diagram ID:** DIA_E01_RAD_TYPES
- **Suggested animation ID:** ANI_E01_HEAT_DISTRIBUTION
- **Suggested print card ID:** CRD_E01_RAD_GUIDE

Focus 17: Warm Radiators in Low-Temperature Systems

- **Concept ID:** CON_E02
- **Plain-English title:** Why Your New Radiators Will Feel Warm, Not Scalding
- **Category:** Radiators, Emitters, and Heat Loss
- **Customer question it answers:** "My new heating system is running, but the radiators only feel lukewarm. Is my system broken?"
- **Why it matters:** Modern low-temperature systems run continuously at lower flow temperatures, meaning radiators feel warm to the touch rather than scalding hot. Managing this expectation is critical to prevent unnecessary service calls and user frustration.
- **Atlas trigger conditions:** Property is transitioning to a low-temperature system (such as a heat pump or condensing boiler), and the customer is used to a legacy high-temperature boiler.
- **Engine facts needed:** Recommended design flow temperature, heat pump COP, and calculated radiator outputs.
- **Customer wording:** "Older boilers heat radiators to a scalding 75^\circ\text{C} in short,

intense bursts. Modern, highly efficient systems run steadily and quietly at a gentle 45^\circ\text{C}. Your radiators will feel warm to the touch, not hot, but they will keep your rooms at your perfect target temperature consistently."

- *One-line summary:* Modern radiators are designed to feel lukewarm and run continuously, keeping your home consistently warm while using far less energy.
- **What you may notice:** Your radiators feel lukewarm to the touch, and your heating system runs quietly and steadily in the background.
- **What stays familiar:** Your rooms remain at your exact target temperature, and your home feels consistently warm and comfortable.
- **What not to worry about:** Lukewarm radiators are a sign of a perfectly designed, highly efficient system, not a breakdown.
- **Common misunderstanding:** "If my radiators aren't scalding hot, my house won't stay warm in freezing weather."
- **Reality:** Scalding radiators are a sign of an inefficient, high-temperature system. Running radiators at lower, lukewarm temperatures continuously is far more efficient and delivers a steady, comfortable warmth.
- **Dangerous oversimplification to avoid:** Saying "radiators don't need to be hot" without verifying that they have been correctly sized to match your home's heat loss.
- **Best analogy families:** Weather/clothing.
- **Where analogy works:** Comparing heating systems to winter clothing. Older boilers are like stepping outside in a massive, heavy winter coat for ten minutes, then coming inside and stripping down. Heat pumps are like wearing a cozy, light thermal fleece all day—you stay perfectly warm without ever feeling overheated.
- **Where analogy breaks:** Thermal clothing does not actively generate heat, whereas a heating system continuously distributes energy.
- **Best graphic:** A comparative diagram. The left side shows an older boiler with scalding radiators (75^\circ\text{C}) cycling on and off. The right side shows a heat pump with warm radiators (45^\circ\text{C}) running continuously and efficiently.
- **Best animation:** An animation showing room warmth. Watch a room cycle between freezing and stuffy with an older high-temperature boiler, then watch a low-temperature system maintain a steady, comfortable green temperature.
- **Best print sheet:** An A5 "Warm Radiator Guide" explaining how low-temperature heating works and why it is highly efficient.
- **Portal treatment:** An interactive diagnostic card. Users enter their room temperature to verify if their system is working perfectly, even if radiators feel lukewarm.
- **PDF treatment:** A dedicated section in the handover pack explaining what to expect from low-temperature heating.
- **Engineer handover treatment:** Hands-on commissioning checks to verify room temperatures match design targets.
- **Technical appendix:** System design standards. Low-temperature systems must be designed to maintain comfortable room temperatures at design flow temperatures under 55^\circ\text{C}, as required by Part L.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the water flow path.
    - *ADHD:* Keep step-by-step system comparisons short and numbered.
    - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
    - *Technophobia:* Reassure the user that the system runs automatically once set.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the

comparative diagrams.
    - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON[span_62](start_span)[span_62](end_span)[span_68](start_span)[span_68](end_span)_E01, CON_H04, CON_I03.
- **Suggested diagram ID:** DIA_E02_LOW_TEMP
- **Suggested animation ID:** ANI_E02_[span_63](start_span)[span_63](end_span)[span_69](start_span)[span_69](end_span)STEADY_WARMTH
- **Suggested print card ID:** CRD_E02_WARM_RADS

Focus 18: Weather/Load Compensation

- **Concept ID:** CON_G01
- **Plain-English title:** Smart Controls: Adapting Your Heating to the Weather
- **Category:** Controls and Customer Behaviour
- **Customer question it answers:** "What is weather compensation, and how does it help my heating system run more efficiently?"
- **Why it matters:** Without smart controls, heating systems run at fixed, high temperatures even on mild days, wasting fuel and wearing out components. Weather and load compensation adjust heating output in real time based on outdoor temperatures, maximizing efficiency.
- **Atlas trigger conditions:** Recommended on all boiler and heat pump systems generated by the Atlas engine, as required by Building Regulations.
- **Engine facts needed:** Heat source type, outdoor temperature sensor configuration, and recommended heating curves.
- **Customer wording:** "Weather compensation is a smart feature that uses an outdoor sensor to monitor the weather in real time. On freezing winter days, it commands your system to run warmer; on mild autumn days, it automatically lowers water temperatures, keeping your home comfortable while using far less energy."
- **One-line summary:** Weather compensation automatically adjusts your heating output based on outdoor temperatures, keeping your home cozy while reducing energy bills.
- **What you may notice:** Your radiators feel slightly cooler on mild days and warmer on freezing days, while your home temperature remains perfectly steady.
- **What stays familiar:** Your thermostat settings remain the same, and your home is kept at your perfect target temperature.
- **What not to worry about:** The system adjusts itself completely automatically in the background, requiring no manual intervention from you.
- **Common misunderstanding:** "I need to manually adjust my boiler temperature settings when the weather changes."
- **Reality:** Weather compensation handles all adjustments automatically, ensuring your system runs at peak efficiency under all weather conditions.
- **Dangerous oversimplification to avoid:** Describing weather compensation as "just a thermometer" without explaining that it uses a calculated heating curve to optimize system efficiency.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing smart controls to cruise control in your car. Instead of

slamming on the accelerator (high temperature) and brakes (turning off), cruise control automatically adjusts engine power to match the road conditions, keeping you safe and saving fuel.
- **Where analogy breaks:** Cruise control manages speed, whereas weather compensation balances heat losses and thermal dynamics.
- **Best graphic:** A diagram showing the inverse relationship between outdoor air temperature and system water flow temperature along a curved slope.
- **Best animation:** An animation of water temperatures. Watch water temperatures rise automatically as snow falls on an outdoor sensor, then watch water temperatures drop smoothly as the sun comes out.
- **Best print sheet:** An A5 "Guide to Smart Controls" explaining weather compensation, load compensation, and how they save energy.
- **Portal treatment:** An interactive graph. Users click on different outdoor temperatures to see how the system automatically adjusts flow temperatures to maximize efficiency.
- **PDF treatment:** A dedicated page in the proposal explaining the smart control design choices.
- **Engineer handover treatment:** Commissioning checks to verify the heating curve and outdoor sensor configurations.
- **Technical appendix:** Smart control requirements. Systems must feature weather or load compensation controls to comply with Boiler Plus and Part L, ensuring flow temperatures are adjusted automatically based on heat demand.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the water flow path.
  - *ADHD:* Keep step-by-step system comparisons short and numbered.
  - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
  - *Technophobia:* Reassure the user that the system runs automatically once set.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_G02, CON_H04, CON_I03.
- **Suggested diagram ID:** DIA_G01_WEATHER_COMP
- **Suggested animation ID:** ANI_G01_COMP_ADJUST
- **Suggested print card ID:** CRD_G01_CONTROL_GDE

Focus 19: Heat Pump Defrost

- **Concept ID:** CON_H01
- **Plain-English title:** Heat Pump Defrosting: Your System's Natural Winter Cycle
- **Category:** Heat Pumps
- **Customer question it answers:** "My heat pump's outdoor unit is blowing out steam and has ice on it. Is it broken?"
- **Why it matters:** In cold, damp winter conditions, frost naturally gathers on the outdoor unit's coils. To maintain efficiency, the heat pump will automatically run a brief "defrost cycle" to melt this ice, occasionally producing harmless steam. Managing this expectation prevents unnecessary service calls.
- **Atlas trigger conditions:** Property is transitioning to an air source heat pump.
- **Engine facts needed:** Defrost cycle frequency, outdoor temperature, and humidity

sensors.

- **Customer wording:** "When it is freezing outside, moisture from the air can naturally freeze on your heat pump's outdoor coils. The system will automatically run a brief 'defrost cycle' (typically taking under ten minutes) to melt this ice, which can occasionally produce a puff of harmless steam and a low hum."
- **One-line summary:** Defrosting is a natural, automatic cycle that keeps your heat pump running efficiently in cold weather by melting frost from outdoor coils.
- **What you may notice:** A temporary change in fan sound, a low hum, a puddle of water under the unit, and a puff of harmless steam.
- **What stays familiar:** Your home remains perfectly warm, and the system resumes heating as soon as the brief defrost cycle completes.
- **What not to worry about:** Frost gathering on coils and puffs of steam are a sign of a perfectly working system, not a breakdown.
- **Common misunderstanding:** "Ice gathering on my heat pump means it is broken or frozen solid."
- **Reality:** Frost is a natural byproduct of drawing heat from cold air. Modern heat pumps are designed with built-in automatic defrost cycles to manage this safely and keep your system running efficiently.
- **Dangerous oversimplification to avoid:** Saying "don't worry about ice" without clarifying that persistent, thick ice blockages that do not melt after several hours can point to a sensor issue that needs to be checked.
- **Best analogy families:** Factual/no analogy.
- **Where analogy works:** Comparing a defrost cycle to your refrigerator's automatic defrost. Both systems melt frost from cooling coils automatically to maintain efficiency, requiring no manual intervention from you.
- **Where analogy breaks:** Refrigerator defrosts run inside your kitchen and do not produce visible steam outside.
- **Best graphic:** A diagram showing frost melting from outdoor coils and draining safely into a waste pipe, with a clear "Harmless Steam" callout.
- **Best animation:** An animation of a defrost cycle. Watch frost gather on outdoor coils, watch the system reverse flow to melt the frost, showing a puff of steam and water draining away, and watch the coils return to clean metal.
- **Best print sheet:** An A5 "Heat Pump Winter Guide" explaining defrosting, ice management, and winter care.
- **Portal treatment:** An interactive diagnostic card. Users enter their observations to verify if their system is running a normal defrost cycle or needs a service visit.
- **PDF treatment:** A dedicated section in the handover pack explaining winter operation.
- **Engineer handover treatment:** Verification of outdoor drainage clearances and a tutorial explaining winter care [CON_H06, CON_I03].
- **Technical appendix:** System design parameters. Defrost cycles reverse the refrigeration cycle, using hot gas to heat the outdoor heat exchanger. The system volume must be sized to ensure enough energy is stored (typically requiring a minimum system volume or a dedicated buffer tank) to support defrosting without cooling the home.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the water flow path.
    - *ADHD:* Keep step-by-step system comparisons short and numbered.
    - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
    - *Technophobia:* Reassure the user that the system runs automatically once set.

    - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
    - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_A02, CON_H04, CON_H06.
- **Suggested diagram ID:** DIA_H01_DEFROST
- **Suggested animation ID:** ANI_H01_MELTING_CYCLE
- **Suggested print card ID:** CRD_H01_WINTER_CARE

Focus 20: Smart Tariff Hot Water

- **Concept ID:** CON_C03
- **Plain-English title:** Heating Your Hot Water with Smart, Cheap Electricity
- **Category:** Hot Water Systems
- **Customer question it answers:** "How do I schedule my hot water cylinder to run during cheap tariff hours without running out of hot water?"
- **Why it matters:** Modern dynamic tariffs offer cheap electricity during off-peak hours (like overnight). Scheduling your hot water cylinder to charge during these periods can significantly lower running costs.
- **Atlas trigger conditions:** Property is installing a smart hot water cylinder or heat pump system, and is on a smart time-of-use or dynamic energy tariff.
- **Engine facts needed:** Cylinder volume (L), dynamic tariff pricing schedules, and heat pump COP.
- **Customer wording:** "Smart tariffs offer cheap electricity when grid demand is low. Scheduling your insulated hot water cylinder to heat up during these cheap, off-peak hours allows you to store powerful hot water cheaply, ready for your morning showers."
- **One-line summary:** Heating your hot water cylinder during cheap, off-peak tariff hours allows you to store hot water cheaply, lowering your energy bills.
- **What you may notice:** The cylinder will heat up silently overnight, and your energy bills will show significantly lower rates during these off-peak hours.
- **What stays familiar:** You enjoy powerful, mains-pressure hot water throughout the day, exactly when you need it.
- **What not to worry about:** Your cylinder is heavily insulated, meaning it loses very little heat and keeps water hot for hours.
- **Common misunderstanding:** "Keeping my cylinder hot continuously is cheaper than heating it in scheduled blocks."
- **Reality:** Heating water in scheduled blocks during cheap, off-peak hours is far cheaper than heating it continuously during high-tariff, peak hours.
- **Dangerous oversimplification to avoid:** Scheduling hot water times without checking that the cylinder is large enough to meet your daily family demand, which can lead to running out of hot water on busy afternoons.
- **Best analogy families:** Money/budget.
- **Where analogy works:** Comparing hot water scheduling to buying fuel at a discount. If a gas station offers fuel at half price overnight, you fill your car's tank fully during those cheap hours, ready for your daily commute. Storing hot water is like filling up at a discount.
- **Where analogy breaks:** Cars burn fuel during driving, whereas hot water cylinders lose a tiny bit of heat slowly over time even when not in use.

- **Best graphic:** A comparative diagram showing off-peak electricity rates and hot water heating schedules side by side.
- **Best animation:** An animation of a clock. Watch electricity rates drop overnight, showing a hot water cylinder heat up and store energy cheaply, ready to deliver powerful hot water during busy peak hours.
- **Best print sheet:** An A5 "Smart Tariff Guide" explaining scheduling, costs, and optimization.
- **Portal treatment:** An interactive graph. Users adjust hot water schedules to see how it affects daily costs and savings.
- **PDF treatment:** A dedicated page in the proposal explaining the recommended tariff and scheduling choices.
- **Engineer handover treatment:** Commissioning checks to verify smart control and scheduling configurations.
- **Technical appendix:** Thermodynamic efficiency metrics. Cylinder standing heat losses must comply with ErP standards, ensuring they do not exceed 1.5\text{ kWh/24h} for a typical 200\text{L} vessel. Legionella pasteurization cycles must be scheduled weekly to maintain hygiene.
- **Accessibility notes:**
  - *Dyslexia:* Use clear visual representations of the water flow path.
  - *ADHD:* Keep step-by-step system comparisons short and numbered.
  - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
  - *Technophobia:* Reassure the user that the system runs automatically once set.
  - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.
  - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_A08, CON_C01, CON_J03.
- **Suggested diagram ID:** DIA_C03_SMART_TARIFF
- **Suggested animation ID:** ANI_C03_TARIFF_SCHED
- **Suggested print card ID:** CRD_C03_TARIFF_GUIDE

Focus 21: What Customers Should Expect Day-to-Day

- **Concept ID:** CON_I01_DAY_TO_DAY
- **Plain-English title:** What to Expect: Day-to-Day Life with Your Modern Heating System
- **Category:** Installation and Handover
- **Customer question it answers:** "How does living with a modern low-temperature system differ from my old high-temperature boiler on a daily basis?"
- **Why it matters:** Shifting customer behavior from a "rapid-blast" mindset to a "gentle-steady" operational model is critical to achieving real-world comfort and efficiency.
- **Atlas trigger conditions:** Handover phase of every low-temperature boiler or heat pump system install.
- **Engine facts needed:** System type (heat pump vs. modulating boiler), recommended setback temperatures, and control interface.
- **Customer wording:** "Older boilers heat homes in short, scalding bursts. Modern, highly efficient systems run gently and steadily in the background. Your radiators will feel lukewarm, and your system will run quietly for longer periods, keeping your home perfectly comfortable while using far less energy."

- **One-line summary:** Modern heating systems provide steady, quiet warmth continuously, requiring very few manual adjustments from you.
- **What you may notice:** Lukewarm radiators, quiet background operation, steady home temperatures, and lower energy bills.
- **What stays familiar:** Your rooms stay at your exact target temperatures, and your home feels consistently cozy.
- **What not to worry about:** Lukewarm radiators and long running times are a sign of a perfectly working, highly efficient system, not a breakdown.
- **Common misunderstanding:** "I need to turn my thermostat up to 30^\circ\text{C} to heat my home up quickly."
- **Reality:** Thermostats act like simple on/off switches, not throttles. Turning them up higher will not heat your home any faster; it will simply overheat your rooms, wasting fuel and increasing bills. Setting a steady target temperature is far more efficient.
- *Dangerous oversimplification to avoid:* Saying "just leave it on all the time" without explaining that a small "setback temperature" (like dropping the temperature by 2\text[span_1065](start_span)[span_1065](end_span)[span_1067](start_span)[span_1067](end_span)[span_1069](start_span)[span_1069](end_span){--}3^\circ\text{C} at night) can keep your home comfortable while maximizing efficiency.
- **Best analogy families:** Car/traffic.
- **Where analogy works:** Comparing heating styles to driving on a highway. Older boilers are like flooring the gas pedal to 100\text{ mph}, slamming on the brakes, stopping, and accelerating again. Modern systems use cruise control to stay at a steady, efficient 60\text{ mph}—you reach your destination comfortably while using far less fuel.
- **Where analogy breaks:** Cruise control manages speed, whereas heating systems balance building heat losses.
- **Best graphic:** A diagram showing daily indoor temperature profiles: an uneven, jagged line with an older boiler vs. a smooth, steady line with a modern modulating system.
- **Best animation:** An animation of room comfort. Watch a room cycle between stuffy and freezing with rapid adjustments, then watch a steady system maintain a smooth, comfortable green temperature continuously.
- **Best print sheet:** An A5 "Day-to-Day Guide" explaining temperatures, setbacks, and basic care.
- **Portal treatment:** An interactive home graphic. Users click on rooms to see how steady, low-temperature heating maintains consistent comfort across their home.
- **PDF treatment:** A dedicated section in the handover pack explaining what to expect from low-temperature heating.
- **Engineer handover treatment:** A physical tutorial explaining controls and setbacks during handover.
- **Technical appendix:** System operation standards. System controls must be configured to support continuous, low-temperature operation, ensuring room temperatures remain stable within +/-0.5^\circ\text{C} of the setpoint.
- **Accessibility notes:**
    - *Dyslexia:* Use clear visual representations of the water flow path.
    - *ADHD:* Keep step-by-step system comparisons short and numbered.
    - *Low literacy:* Focus on the terms "static pressure" and "dynamic flow."
    - *Technophobia:* Reassure the user that the system runs automatically once set.
    - *Visual impairment:* Ensure descriptive alternative text is provided for the comparative diagrams.

    - *Motion sensitivity:* Use simple static animations with clear labels.
- **Priority:** MVP
- **Related concepts:** CON_E02, CON_G02, CON_H04.
- **Suggested diagram ID:** DIA_I01_DAY_LIFE
- **Suggested animation ID:** ANI_I01_STEADY_RUN
- **Suggested print card ID:** CRD_I01_USER_GDE

Section 4: Programmatic Library Implementation Maps

E. Top 25 Concepts to Author First

The authoring roadmap is guided by two factors: clearing technical friction early in the sales cycle and preventing post-installation troubleshooting calls.

Sequence	Concept ID	Domain	Topic	Primary Target Audience
1	CON_F02	Domain F	Why Flushing Protects Your New Boiler	Customer
2	CON_F03	Domain F	When Powerflushing Is Risky	Customer & Engineer
3	CON_F01	Domain F	Powerflush vs. Chemical Flush	Customer
4	CON_B01	Domain B	Boiler Sizing: Why Bigger Is Not Better	Customer & Surveyor
5	CON_B02	Domain B	Gas Pipe Sizing and Pressure Drop	Customer & Engineer
6	CON_D03	Domain D	Microbore Pipe Hydraulic Limitations	Customer & Engineer
7	CON_E01	Domain E	Radiator Surface Area and Output	Customer
8	CON_E02	Domain E	Warm Radiators in Low-Temp Systems	Customer
9	CON_G01	Domain G	Weather and Load Compensation	Customer
10	CON_H01	Domain H	Heat Pump Defrost Cycles and Steam	Customer
11	CON_C01	Domain C	Unvented Cylinder Safety and G3 Code	Customer & Engineer
12	CON_C02	Domain C	Hot Water: Pressure vs.	Customer

Sequence	Concept ID	Domain	Topic	Primary Target Audience
			Storage Volume	
13	CON_D01	Domain D	Static Water Pressure vs. Dynamic Flow	Customer & Surveyor
14	CON_D02	Domain D	Accumulators vs. Booster Pumps	Customer
15	CON_B03	Domain B	System Pressure and the Filling Loop	Customer
16	CON_A01	Domain A	Open-Vented to Sealed Conversion	Customer & Surveyor
17	CON_D04	Domain D	Hydraulic Balancing and Lockshields	Customer & Engineer
18	CON_A02	Domain A	Low-Loss Headers and Buffer Vessels	Engineer & Customer
19	CON_C03	Domain C	Smart Tariff Hot Water Scheduling	Customer
20	CON_I01_DAY_TO_DAY	Domain I	What Customers Should Expect Day-to-Day	Customer
21	CON_E03	Domain E	Room-by-Room Heat Loss Calculations	Customer & Surveyor
22	CON_H04	Domain H	Continuous Low-Temperature Running	Customer
23	CON_F04	Domain F	Magnetic Filters and Sludge Protection	Customer
24	CON_G02	Domain G	Thermostat Basics vs. Dial-Up Fallacies	Customer
25	CON_I02	Domain I	Loft Tank Decommissioning & Space Reclaim	Customer

F. Top 15 Diagrams to Create First

These visual assets are designed to clarify structural transitions, pressure variations, and flow boundaries.

Diagram ID	Concept ID	Visual Focus	Key Structural Elements to Label
DIA_F01_PF_VS_CF	CON_F01	Comparative physical flow loop paths.	Existing system pump path vs. external Powerflush unit path.
DIA_F02_HX_CLOGGING	CON_F02	Cross-section scale comparisons of waterways.	Modern 1.5\text{ mm} coil passages vs. legacy wide cast-iron sections.
DIA_F03_LEAK_RISK	CON_F03	Mechanical rust plug breakdown

inside steel. | Rust blister, active pinhole, static sludge plug, water boundary. | | DIA_F04_MAG_FILTER | CON_F04 | Dynamic magnetic separation flow chamber. | Neodymium core, HPS swirl chamber, collected black magnetite. | | DIA_B01_BOILER_SIZING | CON_B01 | Firing cycle timeline comparison graphs. | On/off short cycling peaks vs. smooth, modulating fuel input curves. | | DIA_B02_GAS_FLOW | CON_[span_611](start_span)[span_611](end_span)[span_632](start_span)[span_632](end_span)B02 | Pipe carrying capacities and friction. | 15\text{ mm} vs. 22\text{ mm} internal area, flow speed, and pressure drops. | | DIA_B03_FILLING_LOOP | CON_B03 | Safe pressure refill step-by-step schematics. | Double-check valves, braided loop, pressure gauge green zone. | | DIA_A01_SEALED_CONV | CON_A01 | Open-vented vs. sealed conversions. | Feed & expansion tank removal, sealed expansion vessel, PRV. | | DIA_C01_CYLINDER_G3 | CON_C01 | G3 safety layout and pipe sizes. | Cylinder, TPRV, D1 metal pipe (<600\text{ mm}), tundish air break, D2 fall. | | DIA_C02_FLOW_COMP | CON_C02 | Flow delivery: combi vs. stored hot water. | Instant heater vs. a balanced unvented mains storage cylinder. | | DIA_D01_PRESSURE_FLOW | CON_D01 | Closed taps (static) vs. running taps (dynamic). | Static pressure gauge reading vs. falling dynamic pressure needle. | | DIA_D02_PRESSURE_BOOST| CON_D02 | Comparative system layouts. | Cold accumulator vessel vs. booster pump set and break tank. | | DIA_D03_MICROBORE | CON_D03 | Friction loss in 10\text{ mm} microbore runs. | Microbore tails, main 22\text{ mm} trunk, high velocity boundary. | | DI[span_21](start_span)[span_21](end_span)[span_39](start_span)[span_39](end_span)A_D[span_65](start_span)[span_65](end_span)[span_71](start_span)[span_71](end_span)04_BALANCING | CON_D04 | Unbalanced vs. balanced radiator circuits. | Shortcut paths of least resistance vs. lockshield balanced flows. | | DIA_G01_WEATHER_COMP | CON_G01 | Inverse relationship: weather heating curve. | Outdoor sensor temperature axis vs. target flow water temperature. |

G. Top 15 Animations to Create First

These digital assets are designed to demonstrate physical processes, fluid dynamics, and automated adjustments in real time.

Animation ID	Concept ID	Animation Behavior	Interactive Loop Control
ANI_F01_PF_VELOCITY	CON_F01	Sludge lifting and clearing under Powerflush velocity.	Hover to tap the radiator and watch sludge sweep away.
ANI_F02_SLUDGE_TRAVE[span_140](start_span)[span_140](end_span)[span_144](start_span)[span_144](end_span)L	CON_F02	Sludge traveling from old radiators and blocking a modern boiler.	Drag a slider to see how older rust behaves in narrow pipes.
ANI_F03_PINHOLE_LEAKS	CON_F03	Sludge dissolving, revealing a pre-existing pinhole leak.	Click to wash sludge away and see where leaks form.
ANI_F04_MAGNET_CAPTURE	CON_F04	Circulating iron particles trapping on a	Drag to pull out the virtual magnet and see

Animation ID	Concept ID	Animation Behavior	Interactive Loop Control
		magnetic rod.	captured rust.
ANI_B01_MODULATION	CON_B01	Modulating burner flame height adjusting to room load.	Click to swap between oversized and modulating systems.
	ANI_B02_PRES[span_658](start_span)[span_658](end_span)SURE_DROP	CON_B02	Gas pressure drop under full load in narrow pipes.
ANI_B03_[span_320](start_span)[span_320](end_span)[span_326](start_span)[span_326](end_span)PRESSURE_REFILL	CO[span_397](start_span)[span_397](end_span)N_[span_354](start_span)[span_354](end_span)[span_369](start_span)[span_369](end_span)B03	Valve adjustments and pressure gauge needle rising.	Click to turn the virtual valves and fill the pressure gauge.
ANI_A01_EXPANSION	CON_A01	Water expanding into a loft tank vs. a sealed vessel.	Tap to start heating and see water volume expand.
ANI_C01_TU[span_612](start_span)[span_612](end_span)[span_633](start_span)[span_633](end_span)NDISH_FLOW	CON_C01	Safe unvented cylinder discharge down a tundish.	Click to simulate an overheat and watch water flow.
ANI_C02_PRESSURE_FLOW	CON_C02	Drop in shower performance: combi vs. unvented cylinder.	Click to open a second tap and watch shower performance.
ANI_D01_DYNAMIC_FLOW	CON_D01	Flow dropping and pressure collapsing as multiple taps are opened.	Click to turn on taps and watch the flow drop.
ANI_D02_ACCUMULATOR	CON_D02	Pressurized bladder contracting during water demand.	Tap to draw water and watch the pressurized bladder contract.
ANI_D03_FLOW_LIMITS	CON_D03	Water velocity rising and creating noise in narrow pipes.	Click to increase velocity and see where noise forms.
ANI_D04_FLOW_BALANCE	CON_D04	Adjusting the nearest lockshield to push water to the furthest room.	Drag to balance lockshield valves and heat every room.
ANI_G01_COMP_ADJUST	CON_G01	Weather compensation curve adjusting flow temperatures.	Drag the thermometer to watch heating curves adjust.

H. Top 15 Print Cards to Create First

- **Physical Format:** Double-sided, heavy-duty A5 cards printed on matte-finish, tear-proof cardstock.
- **Engineering Placement Hook:** Designed with a structural punch-hole and cable-tie to hang directly on the corresponding system components in the plant room (e.g., near the boiler, filling loop, or hot water cylinder).

Print Card ID	Target Concept ID	Front-Side Panel Content	Reverse-Side Panel Content	Physical Installation Point
CRD_F01_FLUSH_COMP	CON_F01	Deep Clean vs. Standard Flush.	Sludge checklist & surveyor diagnostics.	Handed over with the quote booklet.
CRD_F02_WARRANTY	CON_F02	Clean water and warranty protection.	BS 7593 water standards checklist.	Handed over with the quote booklet.
CRD_F03_RISK_ASSESS	CON_F03	Powerflushing old systems safely.	Pre-flush system health disclaimers.	Handed over with the quote booklet.
CRD_F04_FILT[span_733](start_span)[span_733](end_span)ER_G[span_756](start_span)[span_756](end_span)[span_769](start_span)[span_769](end_span)DE	CON_F04	Your system's permanent magnetic shield.	Servicing steps & warranty checklist.	Hung directly on the magnetic filter.
CRD_B01_SIZE_MYTH[span_220](start_span)[span_220](end_span)[span_225](start_span)[span_225](end_span)S	CON_B01	Why bigger boilers are not better.	Home heat loss calculation summary.	Handed over with the quote booklet.
CRD_B[span_253](start_span)[span_253](end_span)02_GAS_UP[span_537](start_span)[span_537](end_span)[span_545](start_span)[span_545](end_span)GRADE	CON_B02	Safe gas pipe carrying capacities.	Upgrade routing and floorboard lifting.	Handed over with the quote booklet.
CRD_B03_REPRESSURIZE	CON_B03	Pressure refill step-by-step panel.	System leak logs & phone numbers.	Hung directly on the filling loop.
CRD_A[span_614](start_span)[span_	CON_A01	Modernizing your system to a sealed	Space reclaimed and freeze safety.	Handed over with the quote booklet.

Print Card ID	Target Concept ID	Front-Side Panel Content	Reverse-Side Panel Content	Physical Installation Point
614](end_span)[span_635](start_span)[span_635](end_span)01_SEALED_SYS		loop.		
CRD_C01_SAFETY_CHECK	CON_C01	Cylinder safety devices & G3 compliance.	Discharge pipe and tundish guide.	Hung directly on the unvented cylinder.
CRD_C02_HOT_WATER	CON_C02	Instant heat vs. stored hot water.	Sizing charts and family occupancies.	Handed over with the quote booklet.
CRD_D01_FLOW_GUIDE	CON_D01	Static pressure vs. dynamic flow rates.	Weir gauge flow test calculations.	Handed over with the survey pack.
CRD_D02_BOOST_COMP	CON_D02	Accumulators vs. booster pumps.	Direct-mains pumping limit regulations.	Handed over with the survey pack.
CRD_D03_MICRO_GUIDE	CON_D03	Heat pumps with microbore pipework.	Water velocity limits and pressure drops.	Handed over with the survey pack.
CRD_D04_BALANCE_GDE	CON_D04	Balancing heat across all rooms.	Adjusting radiator lockshield valves.	Hung directly on the central manifold.
CRD_G01_CONTROL_GDE	CON_G01	Smart controls: weather compensation.	Recommended heating curves.	Hung directly on the thermostat bracket.

I. Misconception-to-Explainer Matrix

This matrix maps common customer fallacies to the specific concept profiles that address them. | Customer Fallacy / Misconception | Engineering Reality | Primary Explainer Concept | Supporting Concepts | | :--- | :--- | :--- | :--- | | "A Powerflush will use high-pressure water that bursts my old copper pipes." | Powerflushing uses high velocity, not high static pressure; leaks only form if pre-existing rust holes are exposed. | CON_F0[span_1113](start_span)[span_1113](end_span)[span_1122](start_span)[span_1122](end_span)3 | CON_F01, CON_F07 | | "A bigger boiler is always better because it heats up my rooms faster." | Radiators limit how fast a room can warm up; oversized boilers short-cycle inefficiently and wear out quickly. | CON_B01 | CON_E01, CON_H04 | | "My old heating worked fine on a 15\text{ mm} pipe, so my new boiler will too." | Modern high-power boilers require 22\text{ mm} pipes to deliver enough fuel and prevent pressure drops under load. | CON_B02 | CON_I05 | | "My boiler pressure has dropped to zero, which means my gas supply is empty." | System pressure refers strictly to the water inside your radiator loop, not your gas supply. | CON_B03 | CON_A01, CON_B1[span_692](start_span)[span_692](end_span)[span_699](start_span)[span_699](end_span)1 | | "Converting to a sealed system will cause my old radiators to leak." | Healthy radiators handle the pressure easily; pre-install surveys identify and replace fragile components before

conversion. | [span_549](start_span)[span_549](end_span)[span_552](start_span)[span_552](end_span)CON_A01 | [span_268](start_span)[span_268](end_span)[span_276](start_span)[span_276](end_span)CON_B03, CO[span_482](start_span)[span_482](end_span)[span_498](start_span)[span_498](end_span)N_[span_566](start_span)[span_566](end_span)[span_577](start_span)[span_577](end_span)F07 | | "Unvented cylinders are highly dangerous and prone to exploding." | Modern unvented cylinders feature triple-redundant safety devices and must be installed by certified experts. | CON_C01 | CON_A05, CON_C13 | | "My water pressure is excellent, so I can easily run three showers at once." | High static pressure does not guarantee dynamic flow; narrow water pipes cause pressure to drop under load. | CON_D01 | CON_C02, CON_D05 | | "I have thin microbore pipes, so I absolutely cannot get a heat pump." | Heat pumps run successfully with microbore pipes if the system is designed carefully with appropriate flow settings. | CON_D03 | CON_A02, CON_E02 | | "My new radiators only feel lukewarm, so my heating system must be broken." | Low-temperature systems distribute a gentle, steady warmth continuously, using far less energy than old blast systems. | CON_E02 | CON_H04, CON_I01_DAY_TO_DAY | | "I need to turn my thermostat up to 30^\circ\text{C} to heat my home up quickly." | Thermostats act like simple on/off switches, not throttles; high settings simply overheat your rooms, wasting fuel. | CON_G02 | CON_G01, CON_I01_DAY_TO_DAY | | "Ice gathering on my heat pump unit means it is broken or frozen solid." | Frost is a natural byproduct of heating; systems melt it automatically using built-in defrost cycles. | CON_H01 | CON_H04, CON_I03 |

J. Customer Anxiety/Trust Map

This map identifies primary customer anxieties during the heating transition, and outlines how Atlas can resolve them.

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| ANXIETY 1:
Physic[span_1045](start_span)[span_1045](end_span)[span_1053](start_sp
an)[span_1053](end_span)[span_1061](start_span)[span_1061](end_span)[s
pan_1160](start_span)[span_1160](end_span)[span_1174](start_span)[span
_1174](end_span)al Disruption and Structural Damage |
| - Root Cause: Fear of lifted floorboards, pipe upgrades, and
wal[span_1161](start_span)[span_1161](end_span)[span_1175](start_span)
[span_1175](end_span)l drilling. |
| - Trust Lever: Detailed spatial plans and explicit route layouts. |
| - Explainer Concept: CON_B02 (Gas Pipe Sizing) |
| - Timing Trigger: Immediately after survey data upload. |
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an)[span_852](end_span)[span_861](start_span)[span_861](end_span)-----
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[span_759](start_span)[span_759](end_span)[span_772](start_span)[span_
772](end_span)    v
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| ANXIETY 2: System Safety and Pressurized
Ves[span_827](start_span)[span_827](end_span)[span_831](start_span)[sp
an_831](end_span)[span_835](start_span)[span_835](end_span)sels |
| - Root Cause: Fear of pressurized water cylinders and leak risks. |
| - Trust
Lev[span_738](start_span)[span_738](end_span)[span_741](start_span)[sp
an_741](end_span)[span_744](start_span)[span_744](end_span)er:
Certification[span_646](start_span)[span_646](end_span)[span_648](star
t_span)[span_648](end_span)s of G3 installers and clear safety
schematics. |
| - Explainer Concept: CON_C01 (Cylinder
Safety[span_358](start_span)[span_358](end_span)[span_373](start_span)
[span_373](end_span)) |
| - Timing Trigger: Included alongside unvented cylinder
quotes.[span_414](start_span)[span_414](end_span)[span_424](start_span
)[span_424](end_span) |
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](start_span)[span_380](end_span)[span_382](start_span)[span_382](end_
span)--------------[span_309](start_span)[span_309](end_span)[span_311
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span)-------[span_525](start_span)[span_525](end_span)[span_527](start
_span)[span_527](end_span)---------------------------[span_465](start_
span)[span_465](end_span)-----------+
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425](end_span)     v
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](end_span)----------[span_582](start_span)[span_582](end_span)[span_5
85](start_span)[span_585](end_span)[span_588](start_span)[span_588](en
d_span)-----------------------------+

| ANXIETY 3: Post-Installation Comfort and
Co[span_221](start_span)[span_221](end_span)[span_226](start_span)[spa
n_226](end_span)ld Rooms |
| - Root Cause: Lukewarm radiators and fear of freezing in cold
weather. |
| - Trust Lever: Low-temperature heating physics and user comfort
profiles. |
| - Explainer Concept: CON_E02 (Warm Radiators) |
| - Timing Trigger: Sent via email three days before installation
starts. |
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| ANXIETY 4: High Energy Bills and Operating Costs |
| - Root Cause: Switching from gas to electric heating under
fluctuating rates.|
| - Trust Lever: Smart tariff scheduling and efficiency COP
multipliers. |
| - Explainer Concept: CON_C03 (Smart Tariff Scheduling) |
| - Timing Trigger: Presented during the final system handover phase.
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```

K. Accessibility Content Checklist

To ensure the Atlas Educational Library is accessible to all users, content creators and designers must follow this programmatic checklist across digital and print formats :

1. Dyslexia and Cognitive Comfort

- Use sans-serif fonts with generous line spacing.
- Avoid large blocks of italicized text or uppercase words in main explanations.
- Place a bold, one-line summary at the very beginning of each concept block to capture the core idea instantly.
- Use high-contrast visual status indicators (e.g., green checkmarks and red caution signs) rather than text-heavy tables.

2. ADHD and Focus Optimization

- Use visual progress bars or step-by-step numbers to help users track their reading progress.
- Integrate interactive sliders or click-to-reveal panels to keep users engaged and focused.
- Break detailed technical specifications into separate tabs or "read more" expansion panels to prevent information overload.

3. Low Literacy and Clarity

- Keep explanations simple and direct, avoiding complex engineering jargon where possible.
- Ground every concept in a simple, everyday analogy (e.g., cars, baking, or blankets) before presenting technical details.
- Ensure all key structural components (such as unvented cylinders or filling loops) are supported by high-quality, labeled illustrations.

4. Technophobia and Resilience

- Clearly emphasize that modern heating systems run completely automatically in the background, requiring no daily adjustments from the user.
- Reassure users that systems feature manual physical controls, and outline exactly what happens during Wi-Fi or power outages.
- Provide laminated physical print cards with simple, icon-led instructions that hang directly on the heating equipment.

5. Visual Impairment

- Use high-contrast colors (minimum contrast ratio of 4.5:1) for all digital interfaces and diagrams.
- Provide descriptive alt-text for all diagrams, explaining the key mechanical components and flow paths clearly.
- Ensure all digital text scales cleanly up to 200\% without breaking the page layout.

6. Motion Sensitivity

- Ensure all animations feature clear play, pause, and stop controls rather than looping continuously.
- Support system preferences for reduced motion, disabling automatic animations for users who are sensitive to movement.

L. Recommended Content Schema Fields for Atlas

This JSON Schema defines the structured fields required in the Atlas content management system. It ensures that every educational module has the data needed to trigger dynamically based on engine calculations.

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "AtlasContentRequirement",
  "type": "object",
  "properties": {
    "concept_id": {
      "type": "string",
      "pattern": "^CON_[A-J][0-9]{2}(_[A-Z0-9_]+)?$",
      "description": "Unique identifier linking the concept to the
Atlas engine."
    },
    "plain_english_title": {
      "type": "string",
      "description": "Simple, clear title of the concept for
customers."
    },
    "category": {
      "type": "string",
```

"enum":
},
"customer_question": {
"type": "string",
"description": "The exact question this concept answers for the customer."
},
"why_it_matters": {
"type": "string",
"description": "Detailed explanation of the engineering and financial impact."
},
"trigger_conditions": {
"type": "array",
"items": { "type": "string" },
"description": "Engine triggers (e.g., 'mains_flow < 12', 'pipe_bore == 10') that call this concept."
},
"engine_facts_needed": {
"type": "array",
"items": { "type": "string" },
"description": "Variables (e.g., system volume, heat loss) required from the thermodynamic model."
},
"customer_wording": {
"type": "strin[span_590](start_span)[span_590](end_span)[span_592](start_span)[span_592](end_span)[span_594](start_span)[span_594](end_span)g",
"description": "The exact, approved wording to present to the customer in quotes."
},
"one_line_summary": {
"type": "string",
"description": "A brief, high-impact summary of the concept."
},
"what_you_may_notice": {
"type": "string",
"description": "The physical signs (sounds, water temperatures) the customer will experience."
},
"what_stays_familiar": {
"type": "string",
"description": "What elements of their system will remain unchanged."
},
"what_not_to_worry_about": {
"type": "string",
"description": "Normal system behavior (steam, long runs) that

```
is not a cause for alarm."
    },
    "misconception_buster": {
      "type": "object",
      "properties": {
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        "reality": { "type": "string" }
      },
      "required": ["misconstanding", "reality"]
    },
    "dangerous_oversimplification": {
      "type": "string",
      "description": "Technical simplifications to avoid to prevent
installer liability."
    },
    "analogy": {
      "type": "object",
      "properties": {
        "family": {
          "type": "string",
          "enum": [
            "water/tank/straw",
            "car/traffic",
            "kitchen/cooking",
            "body/health",
            "money/budget",
            "weather/clothing",
            "factual/no analogy"
          ]
        },
        "works": { "type": "string" },
        "breaks": { "type": "string" }
      },
      "required": ["family", "works", "breaks"]
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    "assets": {
      "type": "object",
      "properties": {
        "diagram_id": { "type": "string", "pattern":
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        "animation_id": { "type": "string", "pattern":
"^ANI_[A-J][0-9]{2}(_[A-Z0-9_]+)?$" },
        "print_card_id": { "type": "string", "pattern":
"^CRD_[A-J][0-9]{2}(_[A-Z0-9_]+)?$" }
      },
      "required": ["diagram_id", "animation_id", "print_card_id"]
    },
    "treatments": {
```

```
      "type": "object",
      "properties": {
        "portal": { "type": "string" },
        "pdf": { "type": "string" },
        "engineer_handover": { "type": "string" },
        "technical_appendix": { "type": "string" }
      },
      "required": ["portal", "pdf", "engineer_handover",
"technical_appendix"]
    },
    "accessibility_notes": {
      "type": "array",
      "items": { "type": "string" },
      "description": "Tailored accessibility guidance for content
creators."
    },
    "priority_tier": {
      "type": "string",
      "enum": ["MVP", "Phase 2", "Later"]
    },
    "related_concepts": {
      "type": "array",
      "items": { "type": "string" }
    }
  },
  "required": [
    "concept_id",
    "plain_english_title",
    "category",
    "customer_question",
    "why_it_matters",
    "trigger_conditions",
    "engine_facts_needed",
    "customer_wording",
    "one_line_summary",
    "what_you_may_notice",
    "what_stays_familiar",
    "what_not_to_worry_about",
    "misconception_buster",
    "dangerous_oversimplification",
    "analogy",
    "assets",
    "treatments",
    "accessibility_notes",
    "priority_tier",
    "related_concepts"
  ]
}
```

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